CHAPTER 8

# Alkenes: Reactions and Synthesis



**FIGURE 8.1 The Spectra fiber used to make the bulletproof vests used by police and military is made of ultra-high-molecular-weight polyethylene, a simple alkene polymer.** (credit: modification of work "US Navy 081028-N-3857R-007 Seabees participate in a chemical, biological and radiological warfare drill" by U.S. Navy photo by Mass Communication Specialist 1st Class Chad Runge/Wikimedia Commons, Public Domain)

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**WHY THIS CHAPTER?** Much of the background needed to understand organic reactions has now been covered, and it's time to begin a systematic description of the major functional groups. In this chapter on alkenes, and in future chapters on other functional groups, we'll discuss a variety of reactions, but try to focus on the general principles and patterns of reactivity that tie organic chemistry together. There are no shortcuts; you have to know the reactions to understand organic and biological chemistry.

Alkene addition reactions occur widely, both in the laboratory and in living organisms. Although we've studied

only the addition of HX thus far, many closely related reactions also take place. In this chapter, we'll see briefly how alkenes are prepared and we'll discuss further examples of alkene addition reactions. Particularly important are the addition of a halogen ($X_2$) to give a 1,2-dihalide, addition of a hypohalous acid (HOX) to give a halohydrin, addition of water to give an alcohol, addition of hydrogen to give an alkane, addition of a single oxygen to give a three-membered cyclic ether called an **epoxide**, and addition of two hydroxyl groups to give a 1,2-diol.

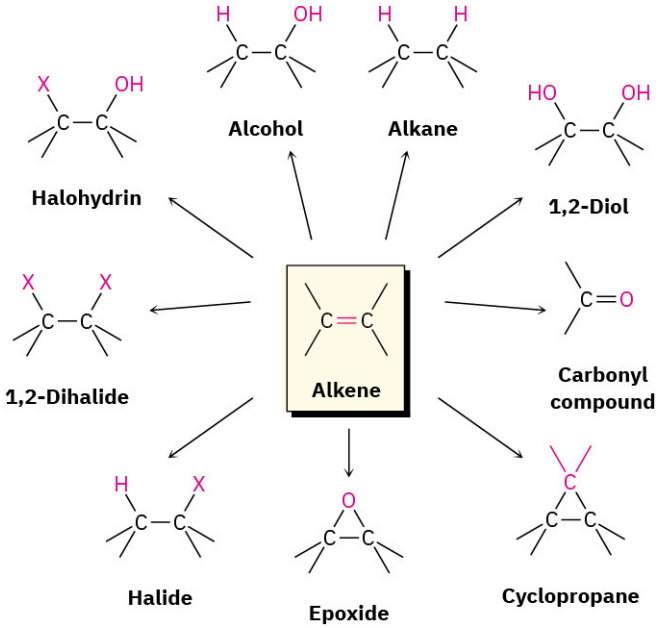
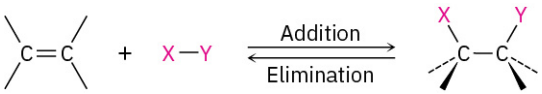



FIGURE 8.2 **Some useful alkene reactions.**

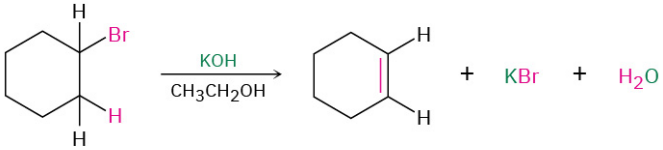
## 8.1 Preparing Alkenes: A Preview of Elimination Reactions

Before getting to the main subject of this chapter—the reactions of alkenes—let's take a brief look at how alkenes are prepared. The subject is a bit complex, though, so we'll return to it in Chapter 11 for a more detailed study. For the present, it's enough to realize that alkenes are readily available from simple precursors—usually alcohols in biological systems and either alcohols or alkyl halides in the laboratory.

Just as the chemistry of alkenes is dominated by addition reactions, the preparation of alkenes is dominated by elimination reactions. Additions and eliminations are, in many respects, two sides of the same coin. That is, an addition reaction might involve the addition of HBr or $H_2O$ to an alkene to form an alkyl halide or alcohol, whereas an elimination reaction might involve the loss of HBr or $H_2O$ from an alkyl halide or alcohol to form an alkene.



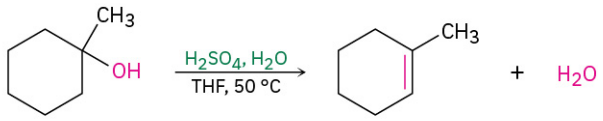

The two most common elimination reactions are dehydrohalogenation—the loss of HX from an alkyl halide—and dehydration—the loss of water from an alcohol. Dehydrohalogenation usually occurs by reaction of an alkyl halide with strong base such as potassium hydroxide. For example, bromocyclohexane yields cyclohexene when treated with KOH in ethanol solution.




**Bromocyclohexane** **Cyclohexene (81%)**

Dehydration is often carried out in the laboratory by treatment of an alcohol with a strong acid. For example, when 1-methylcyclohexanol is warmed with aqueous sulfuric acid in tetrahydrofuran (THF) solvent, loss of

water occurs and 1-methylcyclohexene is formed.

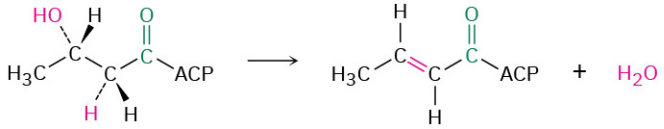



**1-Methylcyclohexanol** **1-Methylcyclohexene (91%)**




**Tetrahydrofuran (THF)—a common solvent**

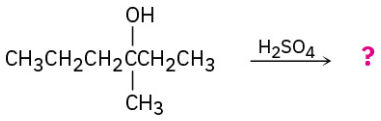
In biological pathways, dehydrations rarely occur with isolated alcohols. Instead, they normally take place on substrates in which the –OH is positioned two carbons away from a C=O group. In the biosynthesis of fats, for instance, *β*-hydroxybutyryl ACP is converted by dehydration to *trans*-crotonyl ACP, where ACP is an abbreviation for *acyl carrier protein.* We'll see the reason for this requirement in **Section 11.10**.




***β*-Hydroxybutyryl ACP** ***trans*-Crotonyl ACP**

**PROBLEM 8-1** One problem with elimination reactions is that mixtures of products are often formed. For example, treatment of 2-bromo-2-methylbutane with KOH in ethanol yields a mixture of two alkene products. What are their likely structures?

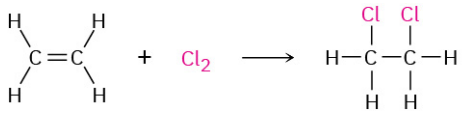
**PROBLEM 8-2** How many alkene products, including *E*,*Z* isomers, might be obtained by dehydration of 3-methyl-3-hexanol with aqueous sulfuric acid?




**3-Methyl-3-hexanol**

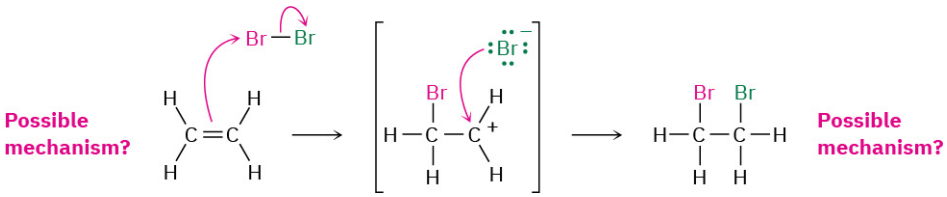
## 8.2 Halogenation of Alkenes: Addition of $X_2$

Bromine and chlorine add rapidly to alkenes to yield 1,2-dihalides, a process called *halogenation*. For example, nearly 50 million tons of 1,2-dichloroethane (ethylene dichloride) are synthesized worldwide each year, much of it by addition of $Cl_2$ to ethylene. The product is used both as a solvent and as starting material for the manufacture of poly(vinyl chloride), PVC, the third most widely synthesized polymer in the world after polyethelyne and polypropolyne. Fluorine is too reactive and difficult to control for most laboratory applications, and iodine does not react with most alkenes.



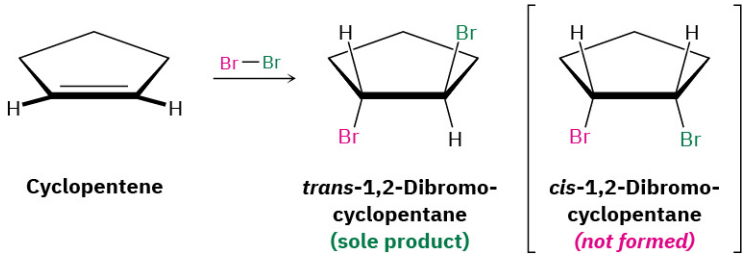

**Ethylene** **1,2-Dichloroethane (ethylene dichloride)**

Based on what we've seen thus far, a possible mechanism for the reaction of bromine with alkenes might involve electrophilic addition of $Br^+$ to the alkene, giving a carbocation intermediate that could undergo further reaction with $Br^-$ to yield the dibromo addition product.

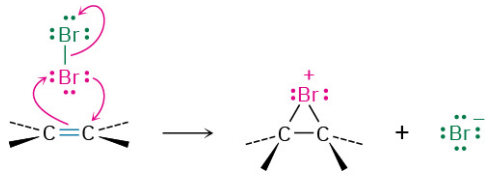



Although this mechanism seems plausible, it's not fully consistent with known facts. In particular, it doesn't explain the stereochemistry of the addition reaction. That is, the mechanism doesn't account for which product stereoisomer is formed.

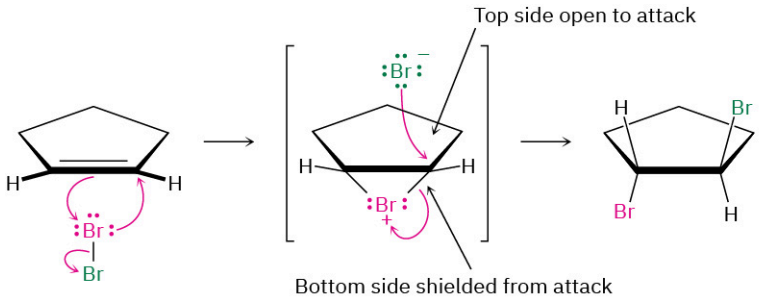
When the halogenation reaction is carried out on a cycloalkene, such as cyclopentene, only the *trans* stereoisomer of the dihalide addition product is formed, rather than the mixture of cis and trans isomers that might have been expected if a planar carbocation intermediate were involved. We say that the reaction occurs with **anti stereochemistry**, meaning that the two bromine atoms come from opposite faces of the double bond—one from the top face and one from the bottom face.

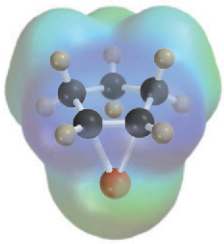



An explanation for the observed stereochemistry of addition was suggested in 1937 by George Kimball and Irving Roberts, who proposed that the reaction intermediate is not a carbocation but is instead a **bromonium ion**, $\mathbf{R_2Br^+}$, formed by electrophilic addition of $Br^+$ to the alkene. (Similarly, a chloronium ion contains a positively charged, divalent chlorine, $R_2Cl^+$.) The bromonium ion is formed in a single step by interaction of the alkene with $Br_2$ and the simultaneous loss of $Br^-$.

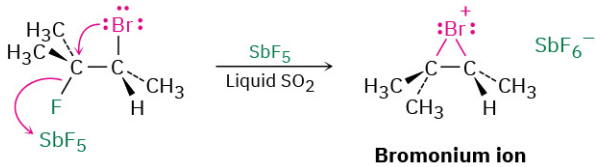



How does the formation of a bromonium ion account for the observed anti stereochemistry of addition to cyclopentene? If a bromonium ion is formed as an intermediate, we can imagine that the large bromine atom might "shield" one side of the molecule. Reaction with $Br^-$ ion in the second step could then occur only from the opposite, unshielded side to give the trans product.



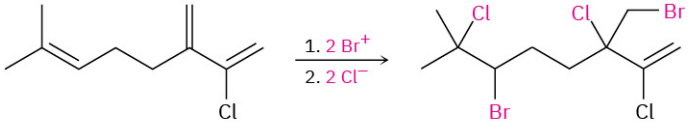



The bromonium ion postulate, made more than 85 years ago to explain the stereochemistry of halogen addition to alkenes, is a remarkable example of deductive logic in chemistry. Arguing from experimental results, chemists were able to make a hypothesis about the intimate mechanistic details of alkene electrophilic reactions. Subsequently, strong evidence supporting the mechanism came from the work of George Olah at the University of Southern California, who prepared and studied stable solutions of cyclic bromonium ions in liquid $SO_2$. There's no question that bromonium ions exist.




**Bromonium ion (stable in $SO_2$ solution)**

Alkene halogenation reactions occur in nature just as they do in the laboratory but are limited primarily to marine organisms living in halide-rich environments. These biological halogenation reactions are carried out by enzymes called *haloperoxidases*, which use $H_2O_2$ to oxidize $Br^-$ or $Cl^-$ ions to a biological equivalent of $Br^+$ or $Cl^+$. Electrophilic addition to the double bond of a substrate molecule then yields a bromonium or chloronium ion intermediate just as in the laboratory, and reaction with another halide ion completes the process. Halomon, for example, an antitumor pentahalide isolated from red alga, is thought to arise by a route that involves twofold addition of BrCl through the corresponding bromonium ions.




**Halomon**

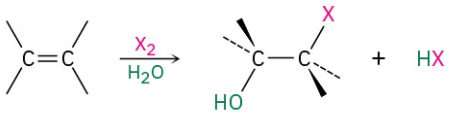
**PROBLEM 8-3** What product would you expect to obtain from addition of $Cl_2$ to 1,2-dimethylcyclohexene? Show the stereochemistry of the product.

**PROBLEM 8-4** Addition of HCl to 1,2-dimethylcyclohexene yields a mixture of two products. Show the stereochemistry of each, and explain why a mixture is formed.

## 8.3 Halohydrins from Alkenes: Addition of HO-X

Another example of an electrophilic addition is the reaction an alkene with either $Br_2$ or $Cl_2$ in the presence of

water to yield a 1,2-halo alcohol, called a **halohydrin**.



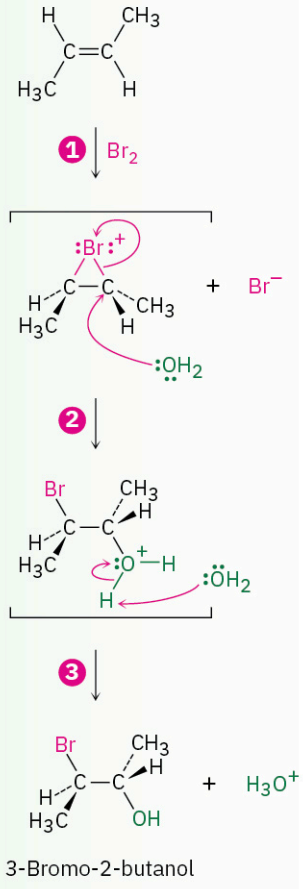

**An alkene** **A halohydrin**

We saw in the previous section that when $Br_2$ reacts with an alkene, the cyclic bromonium ion intermediate reacts with the only nucleophile present, $Br^-$ ion. If the reaction is carried out in the presence of an additional nucleophile, however, the intermediate bromonium ion can be intercepted by the added nucleophile and diverted to a different product. In the presence of a high concentration of water, for instance, water competes with $Br^-$ ion as a nucleophile and reacts with the bromonium ion intermediate to yield a *bromohydrin*. The net effect is addition of HO–Br to the alkene by the pathway shown in FIGURE 8.3.

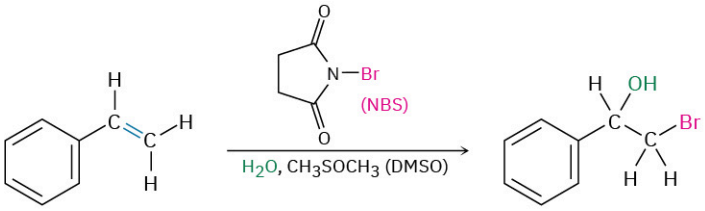
**FIGURE 8.3 MECHANISM**

**Bromohydrin formation by reaction of an alkene with $Br_2$ in the presence of water.**
Water acts as a nucleophile in step 2 to react with the intermediate bromonium ion.

1. Reaction of the alkene with $Br_2$ yields a bromonium ion intermediate, as previously discussed.
2. Water acts as a nucleophile, using a lone pair of electrons to open the bromonium ion ring and form a bond to carbon. Since oxygen donates its electrons in this step, it now has the positive charge.
3. Loss of a proton ($H^+$) from oxygen then gives $H_3O^+$ and the neutral bromohydrin addition product.



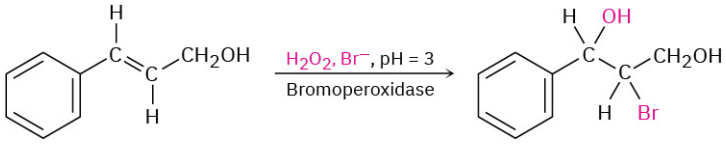

In practice, few alkenes are soluble in water, and bromohydrin formation is often carried out in a solvent such as aqueous dimethyl sulfoxide, $CH_3SOCH_3$ (DMSO), using a reagent called *N*-bromosuccinimide (NBS) as a source of $Br_2$. NBS is a stable, easily handled compound that slowly decomposes in water to yield $Br_2$ at a controlled rate. Bromine itself can also be used in the addition reaction, but it is more dangerous and more difficult to handle than NBS.




**Styrene** **2-Bromo-1-phenylethanol (70%)**

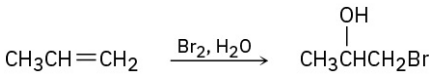
Notice that the aromatic ring in the above example does not react with $Br_2$, even though it appears to have three carbon–carbon double bonds. As we'll see in Section 15.2, aromatic rings are a good deal more stable and less reactive than might be expected.

There are a number of biological examples of halohydrin formation, particularly in marine organisms. As with halogenation (Section 8.2), halohydrin formation is carried out by *haloperoxidases*. For example:




**PROBLEM 8-5** What product would you expect from the reaction of cyclopentene with NBS and water? Show the stereochemistry.

**PROBLEM 8-6** When an unsymmetrical alkene such as propene is treated with *N*-bromosuccinimide in aqueous dimethyl sulfoxide, the major product has the bromine atom bonded to the less highly substituted carbon atom. Is this Markovnikov or non-Markovnikov orientation? (Section 7.8) Explain.

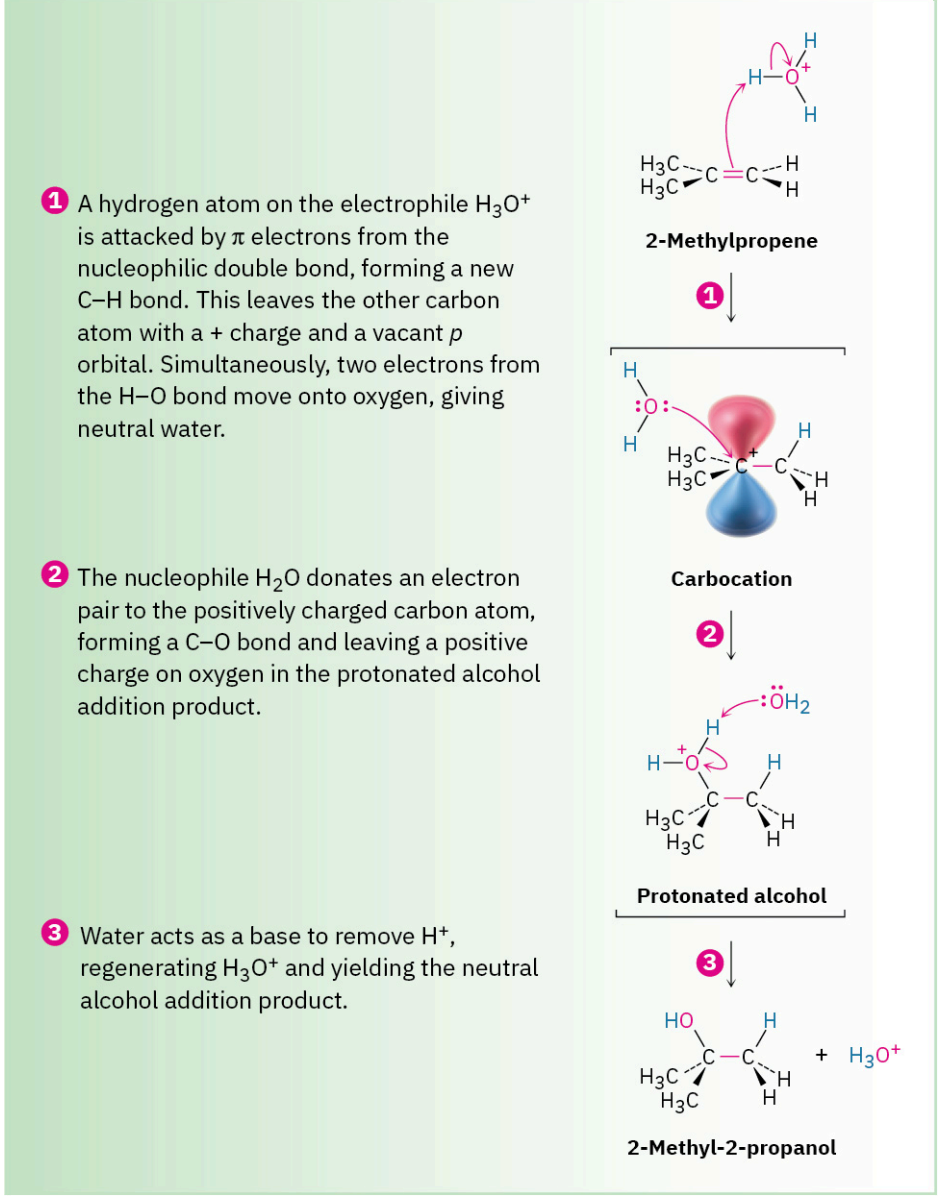



## 8.4 Hydration of Alkenes: Addition of $H_2O$ by Oxymercuration

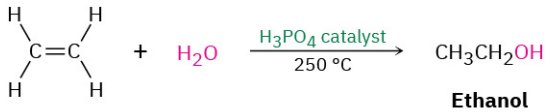
Water adds to alkenes to yield alcohols, a process called hydration. The reaction takes place on treatment of the alkene with water and a strong acid catalyst, such as $H_2SO_4$, by a mechanism similar to that of HX addition. Thus, as shown in FIGURE 8.4, protonation of an alkene double bond yields a carbocation intermediate, which reacts with water to yield a protonated alcohol product, $ROH_2^+$. Loss of $H^+$ from this protonated alcohol gives the neutral alcohol and regenerates the acid catalyst.

**FIGURE 8.4 MECHANISM**

**Mechanism of the acid-catalyzed hydration of an alkene to yield an alcohol.** Protonation of the alkene gives a carbocation intermediate, which reacts with water. The initial product is then deprotonated.



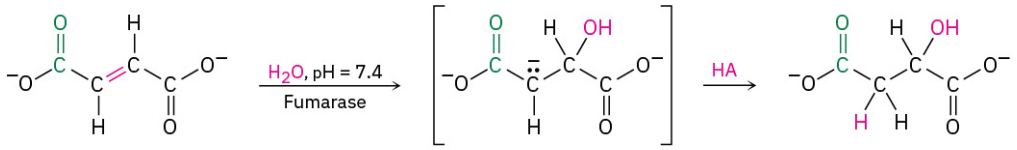

Most ethanol throughout the world is now made by fermentation of biological precursors, such as corn and sugar, but acid-catalyzed alkene hydration is particularly suited to large-scale industrial procedures, and approximately 90,000 tons of ethanol is manufactured each year in the United States by hydration of ethylene. The reaction is of little value in the laboratory, however, because it requires high temperatures—250 °C in the case of ethylene—and strongly acidic conditions.




**Ethylene** **Ethanol**

Acid-catalyzed hydration of double bonds is also uncommon in biological pathways. Instead, biological

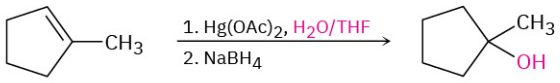
hydrations usually require that the double bond be adjacent to a carbonyl group for the reaction to proceed. Fumarate, for instance, is hydrated to give malate as one step in the citric acid cycle of food metabolism. Note that the requirement for an adjacent carbonyl group in the addition of water is the same as in Section 8.1 for the elimination of water. We'll see the reason for this requirement in **Section 19.13**, but will note for now that the reaction is not an electrophilic addition but instead occurs through a mechanism that involves formation of an anion intermediate followed by protonation by an acid HA.




**Fumarate** **Anion intermediate** **Malate**

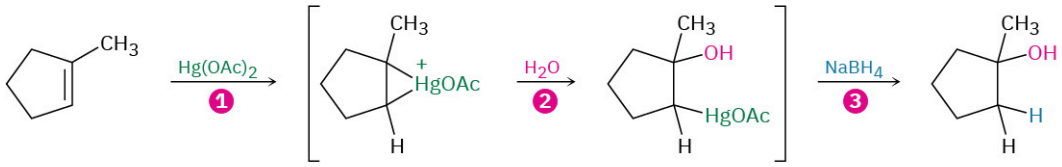
When it comes to circumventing problems like those with acid-catalyzed alkene hydrations, laboratory chemists have a great advantage over the cellular "chemists" in living organisms. Laboratory chemists are not constrained to carry out their reactions in water solution; they can choose from any of a large number of solvents. Laboratory reactions don't need to be carried out at a fixed temperature; they can take place over a wide range of temperatures. And laboratory reagents aren't limited to containing carbon, oxygen, nitrogen, and a few other elements; they can contain any element in the periodic table.

In the laboratory, alkenes are often hydrated by the **oxymercuration–demercuration** procedure, which involves electrophilic addition of $Hg^{2+}$ to the alkene on reaction with mercury(II) acetate $[(CH_3CO_2)_2Hg$. The intermediate organomercury compound is then treated with sodium borohydride, $NaBH_4$, and demercuration occurs to produce an alcohol. For example:




**1-Methylcyclopentene** **1-Methylcyclopentanol (92%)**

Alkene oxymercuration is closely analogous to halohydrin formation. The reaction is initiated by electrophilic addition of $Hg^{2+}$ (mercuric) ion to the alkene to give an intermediate *mercurinium ion*, whose structure resembles that of a bromonium ion (FIGURE 8.5). Nucleophilic addition of water as in halohydrin formation, followed by the loss of a proton, then yields a stable organomercury product. The final step, demercuration of the organomercury compound by reaction with sodium borohydride, is complex and involves radicals. Note that the regiochemistry of the reaction corresponds to Markovnikov addition of water; that is, the –OH group attaches to the more highly substituted carbon atom, and the –H attaches to the less highly substituted carbon. The hydrogen that replaces mercury in the demercuration step can attach from either side of the molecule depending on the exact circumstances.




**1-Methyl-cyclopentene** **Mercurinium ion** **Organomercury compound** **1-Methyl-cyclopentanol (92% yield)**

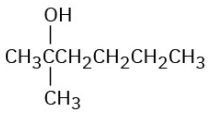
FIGURE 8.5 **Mechanism of the oxymercuration of an alkene to yield an alcohol.** (1) Electrophilic addition of $Hg^{2+}$ gives a mercurinium ion, which (2) reacts with water as in halohydrin formation. Loss of a proton gives an organomercury product, and (3) reaction with $NaBH_4$ removes the mercury. The product of the reaction is a more highly substituted alcohol, corresponding to Markovnikov regiochemistry.

**PROBLEM 8-7** What products would you expect from oxymercuration–demercuration of the following alkenes?

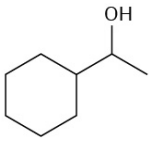
**(a)** $CH_3CH_2CH_2CH{=}CH_2$ **(b)** $CH_3C(CH_3){=}CHCH_2CH_3$

**PROBLEM 8-8** From what alkenes might the following alcohols have been prepared?

**(a)**

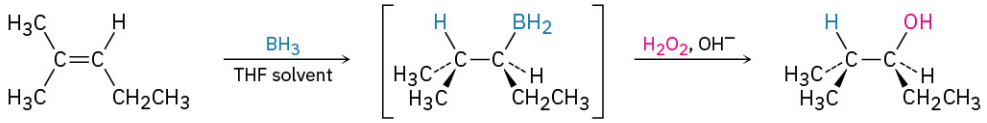



**(b)**



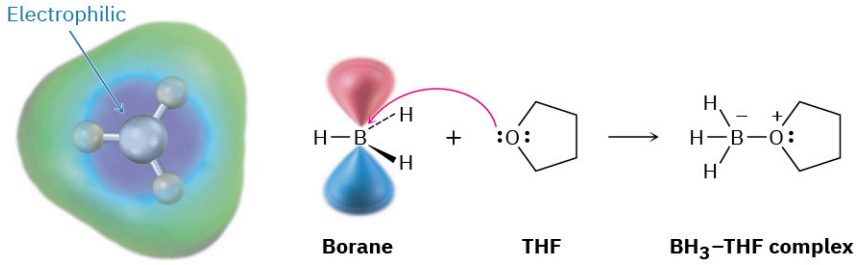

## 8.5 Hydration of Alkenes: Addition of $H_2O$ by Hydroboration

In addition to the oxymercuration–demercuration method, which yields the Markovnikov product, a complementary method that yields the non-Markovnikov product is also useful. Discovered in 1959 by H.C. Brown at Purdue University and called **hydroboration**, the reaction involves addition of a B–H bond of borane, $BH_3$, to an alkene to yield an organoborane intermediate, $RBH_2$. Oxidation of the organoborane by reaction with basic hydrogen peroxide, $H_2O_2$, then gives an alcohol. For example:



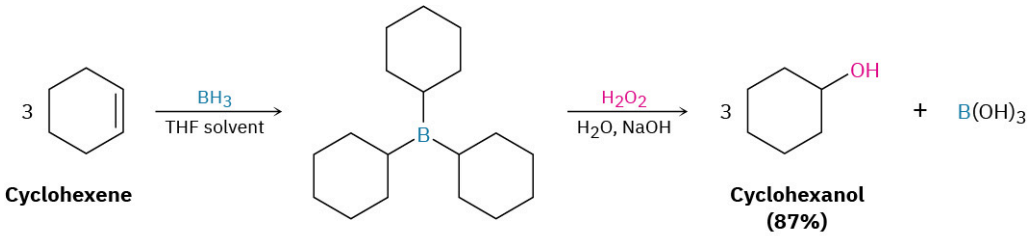

**2-Methyl-2-pentene** | **Organoborane intermediate** | **2-Methyl-3-pentanol**

Borane is very reactive as a Lewis acid because the boron atom has only six electrons in its valence shell rather than an octet. In tetrahydrofuran solution, $BH_3$ accepts an electron pair from a solvent molecule in a Lewis acid–base reaction to complete its octet and form a stable $BH_3$–THF complex.



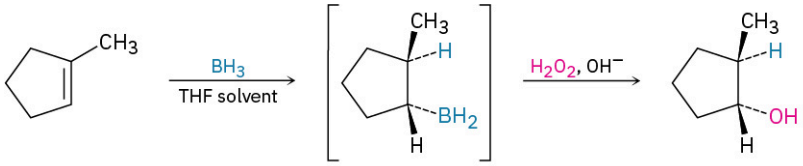

**Borane** | **THF** | **$BH_3$–THF complex**

When an alkene reacts with $BH_3$ in THF solution, rapid addition to the double bond occurs three times and a trialkylborane, $R_3B$, is formed. For example, 1 molar equivalent of $BH_3$ adds to 3 molar equivalents of cyclohexene to yield tricyclohexylborane. When tricyclohexylborane is then treated with aqueous hydrogen $H_2O_2$ in basic solution, an oxidation takes place. The three C–B bonds are broken, –OH groups bond to the three carbons, and 3 equivalents of cyclohexanol are produced. The net effect of the two-step hydroboration–oxidation sequence is hydration of the alkene double bond.




**Cyclohexene** | **Tricyclohexylborane** | **Cyclohexanol (87%)**

One of the features that makes the hydroboration reaction so useful is the regiochemistry that results when an unsymmetrical alkene is hydroborated. For example, hydroboration–oxidation of 1-methylcyclopentene yields *trans*-2-methylcyclopentanol. In this process, boron and hydrogen add to the alkene from the same face of the double bond—that is, with **syn stereochemistry**, the opposite of anti—with boron attaching to the less highly substituted carbon. During the oxidation step, the boron is replaced by an –OH with the same stereochemistry, resulting in an overall syn non-Markovnikov addition of water. This stereochemical result is particularly useful because it is complementary to the Markovnikov regiochemistry observed for oxymercuration–demercuration.




**1-Methyl-cyclopentene** | **Organoborane intermediate** | ***trans*-2-Methyl-cyclopentanol (85% yield)**

Why does alkene hydroboration take place with syn, non-Markovnikov regiochemistry to yield the less highly substituted alcohol? Hydroboration differs from many other alkene addition reactions in that it occurs in a single step without a carbocation intermediate (**FIGURE 8.6**). Because the C–H and C–B bonds form at the same time and from the same face of the alkene, syn stereochemistry results. Non-Markovnikov regiochemistry occurs because attachment of boron is favored at the less sterically crowded carbon atom of the alkene.

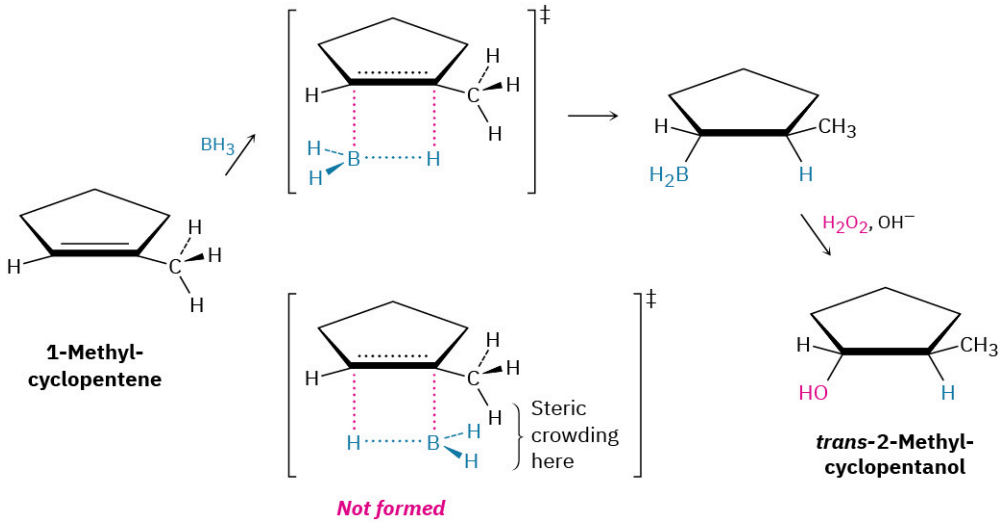



**FIGURE 8.6 Mechanism of alkene hydroboration.** The reaction occurs in a single step in which the C–H and C–B bonds form at the same time and on the same face of the double bond. The lower energy, more rapidly formed transition state is the one with less steric crowding, leading to non-Markovnikov regiochemistry.

## WORKED EXAMPLE 8.1

### Predicting the Products of a Hydration Reaction

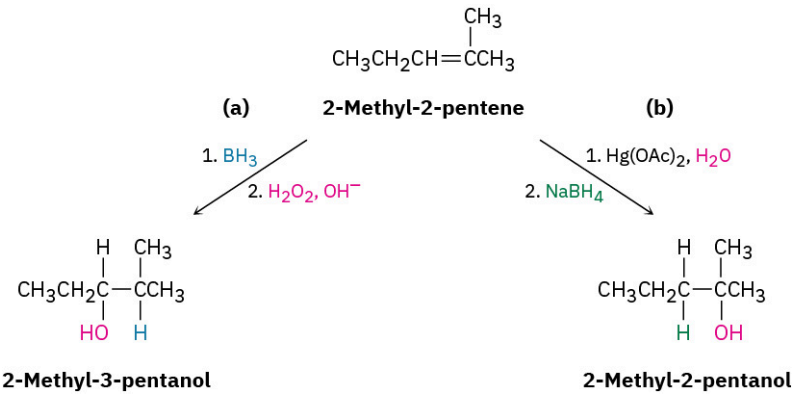
What products would you obtain from reaction of 2-methyl-2-pentene with:

**(a)** $BH_3$, followed by $H_2O_2$, $OH^-$ **(b)** $Hg(OAc)_2$, followed by $NaBH_4$

#### Strategy

When predicting the product of a reaction, you have to recall what you know about the kind of reaction being carried out and apply that knowledge to the specific case you're dealing with. In the present instance, recall that the two methods of hydration—hydroboration–oxidation and oxymercuration–demercuration—give complementary products. Hydroboration–oxidation occurs with syn stereochemistry and gives the non-Markovnikov addition product; oxymercuration–demercuration gives the Markovnikov product.

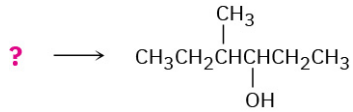
**Solution**




## WORKED EXAMPLE 8.2

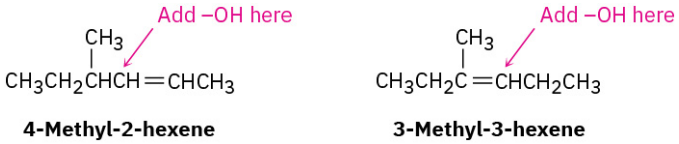
### Synthesizing an Alcohol

How might you prepare the following alcohol?



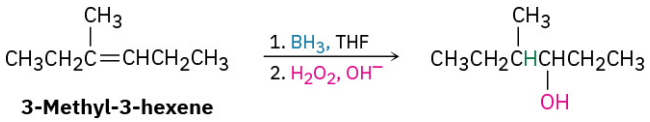

**Strategy**

Problems that require the synthesis of a specific target molecule should always be worked backward. Look at the target, identify its functional group(s), and ask yourself, "What are the methods for preparing that functional group?" In the present instance, the target molecule is a secondary alcohol ($R_2CHOH$), and we've seen that alcohols can be prepared from alkenes by either hydroboration–oxidation or oxymercuration–demercuration. The –OH-bearing carbon in the product must have been a double-bond carbon in the alkene reactant, so there are two possibilities here: 4-methyl-2-hexene and 3-methyl-3-hexene.



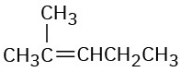

4-Methyl-2-hexene has a disubstituted double bond, RCH=CHR′, and will probably give a mixture of two alcohols with either hydration method since Markovnikov's rule does not apply to symmetrically substituted alkenes. 3-Methyl-3-hexene, however, has a trisubstituted double bond, and should give only the desired product on non-Markovnikov hydration using the hydroboration–oxidation method.

**Solution**

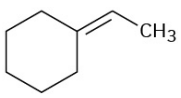



**PROBLEM 8-9** Show the structures of the products you would obtain by hydroboration–oxidation of the following alkenes:

**(a)**



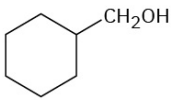

**(b)**



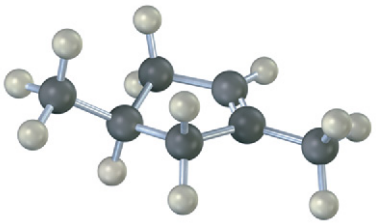
**PROBLEM 8-10** What alkenes might be used to prepare the following alcohols by hydroboration–oxidation?

**(a)** $CH_3CHCH_2CH_2OH$ (with $CH_3$ branch)

**(b)** $CH_3CHCHCH_3$ (with $H_3C$ and OH branches)

**(c)** 


**PROBLEM 8-11** The following cycloalkene gives a mixture of two alcohols on hydroboration followed by oxidation. Draw the structures of both, and explain the result.



## 8.6 Reduction of Alkenes: Hydrogenation

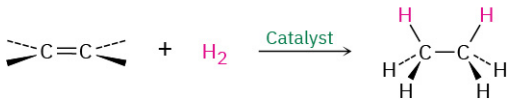
Alkenes react with $H_2$ in the presence of a metal catalyst such as palladium or platinum to yield the corresponding saturated alkanes. We describe the result by saying that the double bond has been **hydrogenated**, or *reduced*. Note that the word *reduction* is used somewhat differently in organic chemistry from what you might have learned previously. In general chemistry, a reduction is defined as the gain of one or more electrons by an atom. In organic chemistry, however, a **reduction** is a reaction that results in a gain of electron density for carbon, caused either by bond formation between carbon and a less electronegative atom—usually hydrogen—or by bond-breaking between carbon and a more electronegative atom—usually oxygen, nitrogen, or a halogen. We'll explore this topic in more detail in **Section 10.8**.

**Reduction** Increases electron density on carbon by:

– forming this: C–H

– or breaking one of these: C–O C–N C–X

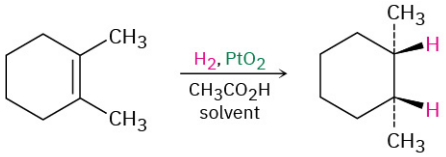
**A reduction:**




**An alkene** **An alkane**

Platinum and palladium are the most common laboratory catalysts for alkene hydrogenations. Palladium is normally used as a very fine powder "supported" on an inert material such as charcoal (Pd/C) to maximize surface area. Platinum is normally used as $PtO_2$, a reagent known as *Adams' catalyst* after its discoverer, Roger Adams at the University of Illinois.

Catalytic hydrogenation, unlike most other organic reactions, is a *heterogeneous* process rather than a homogeneous one. That is, the hydrogenation reaction does not occur in a homogeneous solution but instead takes place on the surface of solid catalyst particles. Hydrogenation usually occurs with syn stereochemistry: both hydrogens add to the double bond from the same face.



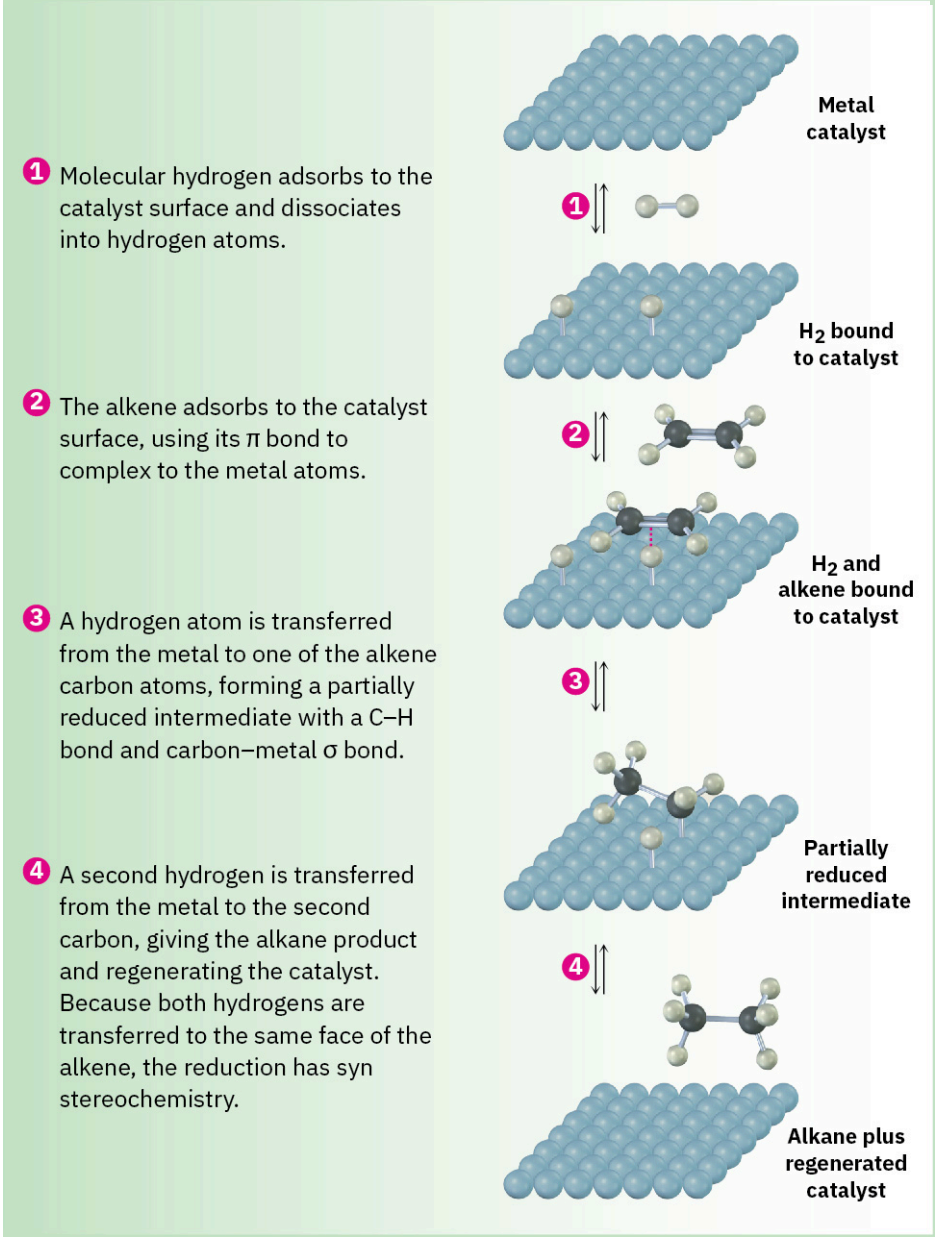

**1,2-Dimethylcyclohexene** ***cis*-1,2-Dimethylcyclohexane (82%)**

As shown in **FIGURE 8.7**, hydrogenation begins with adsorption of $H_2$ onto the catalyst surface. Complexation between catalyst and alkene then occurs as a vacant orbital on the metal interacts with the filled alkene $\pi$ orbital

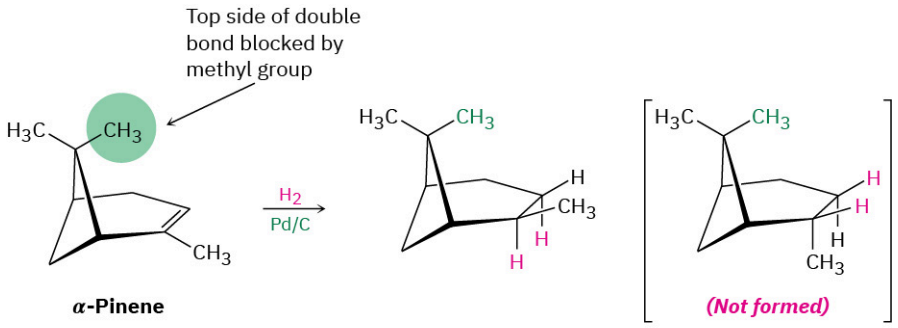
on the alkene. In the final steps, hydrogen is inserted into the double bond and the saturated product diffuses away from the catalyst. The stereochemistry of hydrogenation is syn because both hydrogens add to the double bond from the same catalyst surface.

**FIGURE 8.7 MECHANISM**

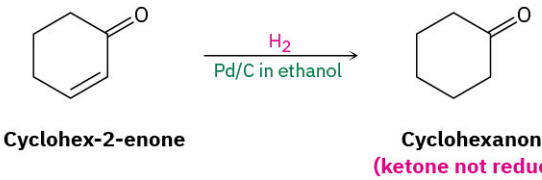
**Mechanism of alkene hydrogenation.** The reaction takes place with syn stereochemistry on the surface of insoluble catalyst particles.



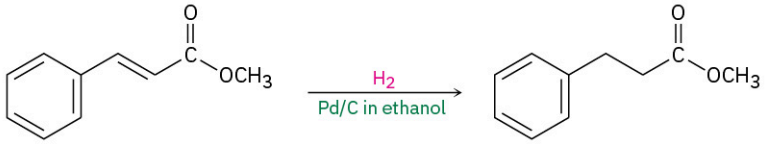

An interesting feature of catalytic hydrogenation is that the reaction is extremely sensitive to the steric environment around the double bond. As a result, the catalyst usually approaches the more accessible face of an alkene, giving rise to a single product. In $\alpha$-pinene, for example, one of the methyl groups attached to the four-membered ring hangs over the top face of the double bond and blocks approach of the hydrogenation catalyst from that side. Reduction therefore occurs exclusively from the bottom face to yield the product shown.



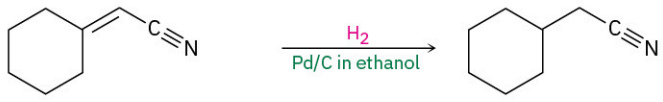

Alkenes are much more reactive toward catalytic hydrogenation than most other unsaturated functional groups, and the reaction is therefore quite selective. Other functional groups, such as aldehydes, ketones, esters, and nitriles, often survive alkene hydrogenation conditions unchanged, although reaction with these groups does occur under more vigorous conditions. Note that, particularly in the hydrogenation of methyl 3-phenylpropenoate shown below, the aromatic ring is not reduced by hydrogen and palladium even though it contains apparent double bonds.




**Cyclohex-2-enone** **Cyclohexanone**
**(ketone not reduced)**




**Methyl 3-phenylpropenoate** **Methyl 3-phenylpropanoate**
**(aromatic ring not reduced)**




**Cyclohexylideneacetonitrile** **Cyclohexylacetonitrile**
**(nitrile not reduced)**

In addition to its usefulness in the laboratory, catalytic hydrogenation is also important in the food industry, where unsaturated vegetable oils are reduced on a large scale to produce the saturated fats used in margarine and cooking products (**FIGURE 8.8**). As we'll see in **Section 27.1**, vegetable oils are triesters of glycerol, $HOCH_2CH(OH)CH_2OH$, with three long-chain carboxylic acids called *fatty acids*. The fatty acids are generally polyunsaturated, and their double bonds have cis stereochemistry. Complete hydrogenation yields the corresponding saturated fatty acids, but incomplete hydrogenation often results in partial cis–trans isomerization of a remaining double bond. When eaten and digested, the free trans fatty acids are released, raising blood cholesterol levels and potentially contributing to coronary problems.

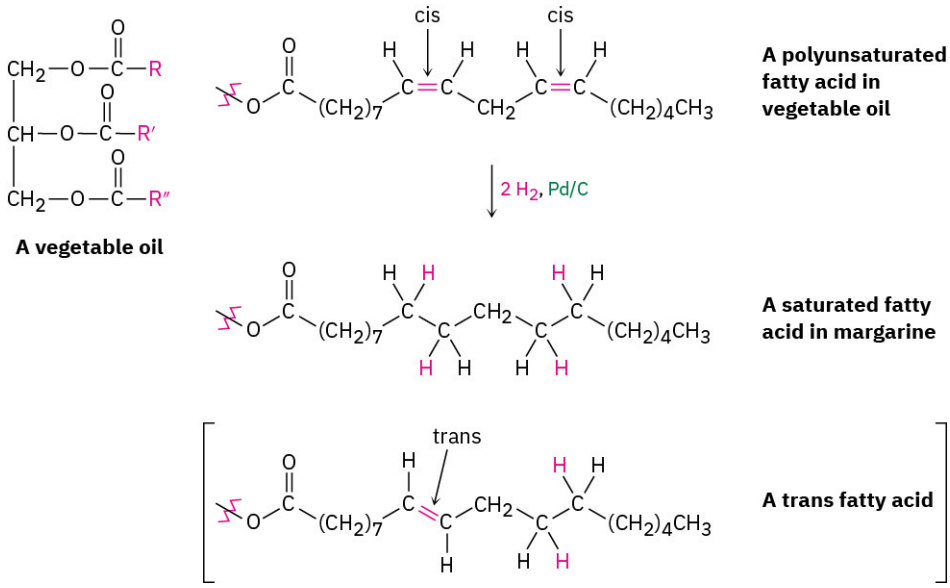



**FIGURE 8.8 Catalytic hydrogenation of polyunsaturated fats leads primarily to saturated products**, along with a small amount of isomerized trans fats.

Double-bond reductions are very common in biological pathways, although the mechanism is completely different from that of laboratory catalytic hydrogenation over palladium. As with biological hydrations (Section 8.4), biological reductions usually occur in two steps and require that the double bond be adjacent to a carbonyl group. In the first step, the biological reducing agent NADPH (reduced nicotinamide adenine dinucleotide phosphate), adds a hydride ion ($H:^-$) to the double bond to give an anion. In the second, the anion is protonated by acid HA, leading to overall addition of $H_2$. An example is the reduction of *trans*-crotonyl ACP to yield butyryl ACP, a step involved in the biosynthesis of fatty acids (FIGURE 8.9).

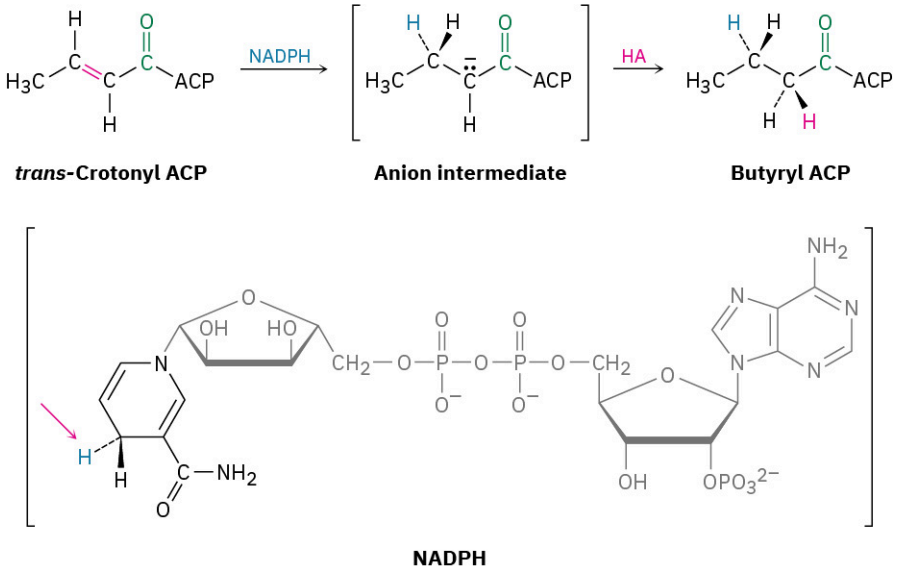
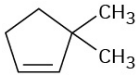
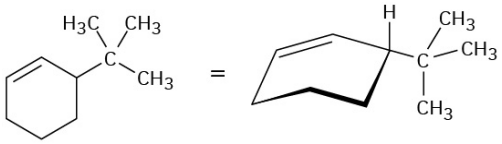



**FIGURE 8.9 Reduction of the carbon–carbon double bond in *trans*-crotonyl ACP, a step in the biosynthesis of fatty acids.** One hydrogen is delivered from NADPH as a hydride ion, $H:^-$; the other hydrogen is delivered by protonation of the anion intermediate with an acid, HA.

**PROBLEM 8-12** What product would you obtain from catalytic hydrogenation of the following alkenes?

**(a)** $CH_3CH{=}CHCH_2CH_3$ **(b)**  **(c)** 

## 8.7 Oxidation of Alkenes: Epoxidation and Hydroxylation

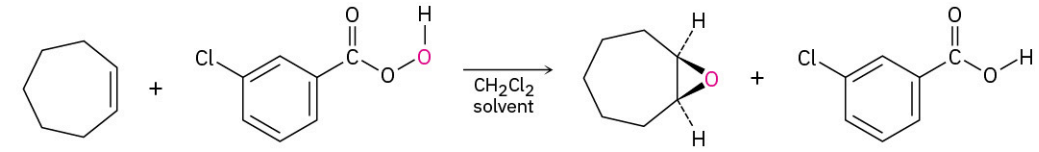
Like the word *reduction* used in the previous section for the addition of hydrogen to a double bond, the word *oxidation* has a slightly different meaning in organic chemistry than what you might have previously learned. In general chemistry, an oxidation is defined as the loss of one or more electrons by an atom. In organic chemistry, however, an **oxidation** is a reaction that results in a loss of electron density for carbon, caused either by bond formation between carbon and a more electronegative atom—usually oxygen, nitrogen, or a halogen—or by bond-breaking between carbon and a less electronegative atom—usually hydrogen. Note that an *oxidation* often adds oxygen, while a *reduction* often adds hydrogen.

**Oxidation** Decreases electron density on carbon by:

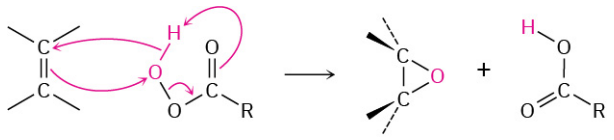
– forming one of these: C–O C–N C–X

– or breaking this: C–H

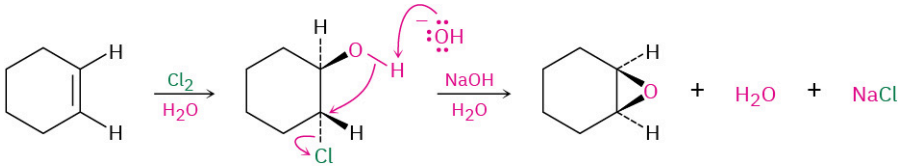
In the laboratory, alkenes are oxidized to give *epoxides* on treatment with a *peroxyacid*, $RCO_3H$, such as *meta*-chloroperoxybenzoic acid. An **epoxide**, also called an **oxirane**, is a cyclic ether with an oxygen atom in a three-membered ring. For example:



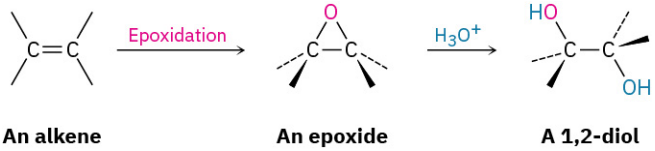

Peroxyacids transfer an oxygen atom to the alkene with syn stereochemistry—both C–O bonds form on the same face of the double bond—through a one-step mechanism without intermediates. The oxygen atom farthest from the carbonyl group is the one transferred.



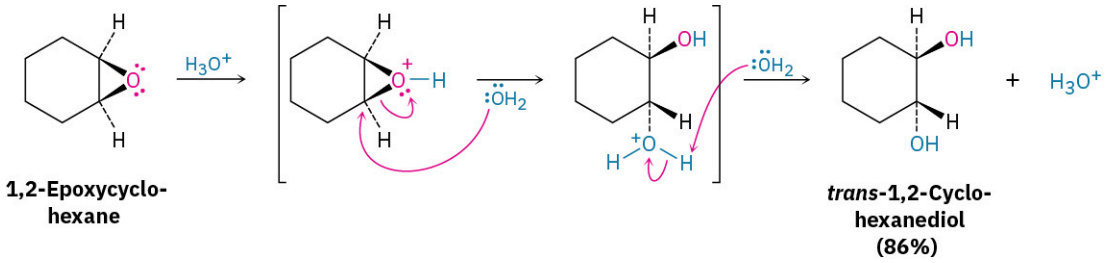

Another method for the synthesis of epoxides involves the use of halohydrins, prepared by electrophilic addition of HO–X to alkenes (Section 8.3). When a halohydrin is treated with base, HX is eliminated and an epoxide is produced.




Epoxides undergo an acid-catalyzed ring-opening reaction with water (a hydrolysis) to give the corresponding 1,2-dialcohol, or *diol*, also called a **glycol**. Thus, the net result of the two-step alkene epoxidation/hydrolysis is **hydroxylation**—the addition of an –OH group to each of the two double-bond carbons. In fact, approximately 204 million tons of ethylene glycol, $HOCH_2CH_2OH$, most of it used for automobile antifreeze, are produced worldwide each year by the epoxidation of ethylene and subsequent hydrolysis.

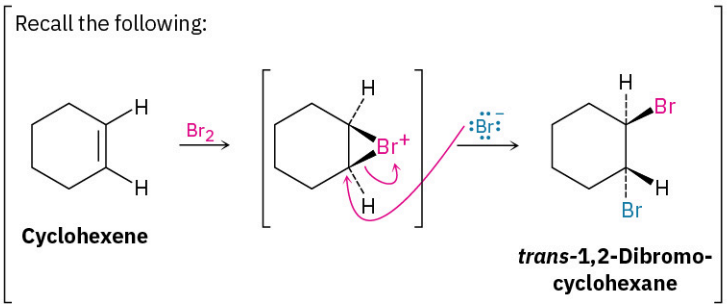


Acid-catalyzed epoxide opening begins with protonation of the epoxide to increase its reactivity, followed by nucleophilic addition of water. This nucleophilic addition is analogous to the final step of alkene bromination, in which a cyclic bromonium ion is opened by a nucleophile (Section 8.2). That is, a *trans*-1,2-diol results when an epoxycycloalkane is opened by aqueous acid, just as a *trans*-1,2-dibromide results when a cycloalkene is brominated. We'll look at epoxide chemistry in more detail in **Section 18.6**.




**1,2-Epoxycyclohexane**

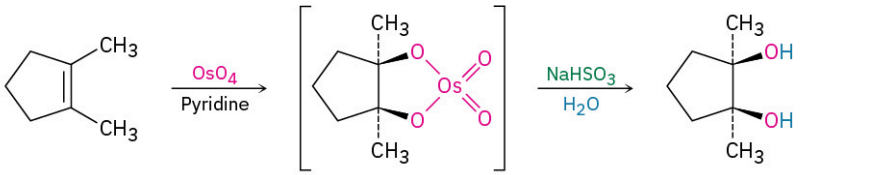
***trans*-1,2-Cyclohexanediol (86%)**




**Cyclohexene**

***trans*-1,2-Dibromocyclohexane**

Hydroxylation can also be carried out directly (without going through an intermediate epoxide) by treating an alkene with osmium tetroxide, $OsO_4$. The reaction occurs with syn stereochemistry and does not involve a carbocation intermediate. Instead, it takes place through an intermediate cyclic *osmate*, which is formed in a single step by addition of $OsO_4$ to the alkene. This cyclic osmate is then cleaved using aqueous sodium bisulfite, $NaHSO_3$.

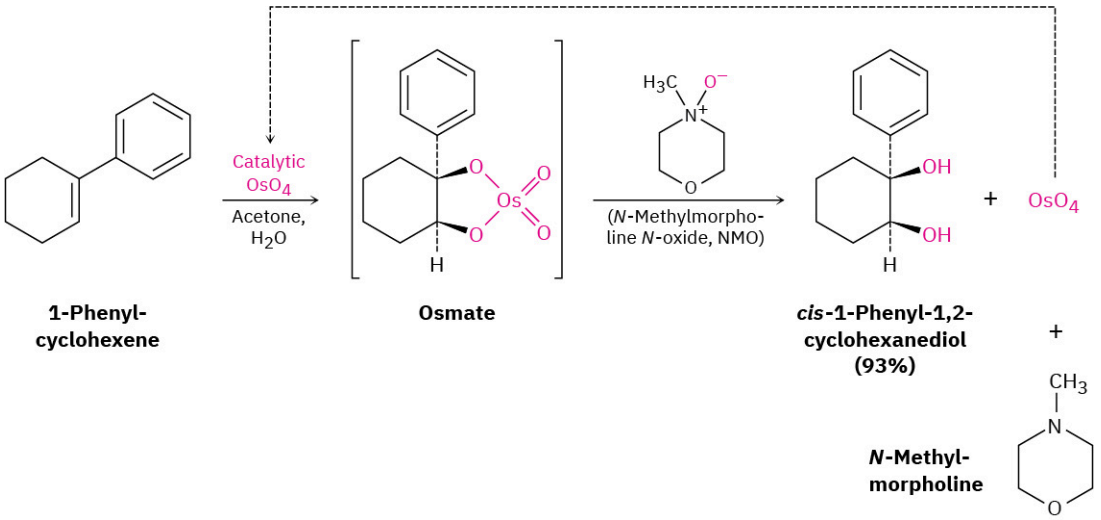



**1,2-Dimethylcyclopentene**

**A cyclic osmate intermediate**

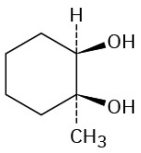
***cis*-1,2-Dimethyl-1,2-cyclopentanediol (87%)**

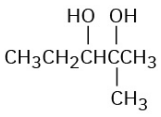
Because $OsO_4$ is both very expensive and *very* toxic, the reaction is usually carried out using only a small, catalytic amount of $OsO_4$ in the presence of a stoichiometric amount of a safe and inexpensive co-oxidant such as *N*-methylmorpholine *N*-oxide, abbreviated NMO. The initially formed osmate intermediate reacts rapidly with NMO to yield the product diol plus *N*-methylmorpholine and reoxidized $OsO_4$, which reacts with more alkene in a catalytic cycle.

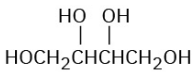



**PROBLEM 8-13** What product would you expect from reaction of *cis*-2-butene with *meta*-chloroperoxybenzoic acid? Show the stereochemistry.

**PROBLEM 8-14** Starting with an alkene, how would you prepare each of the following compounds?

**(a)** 

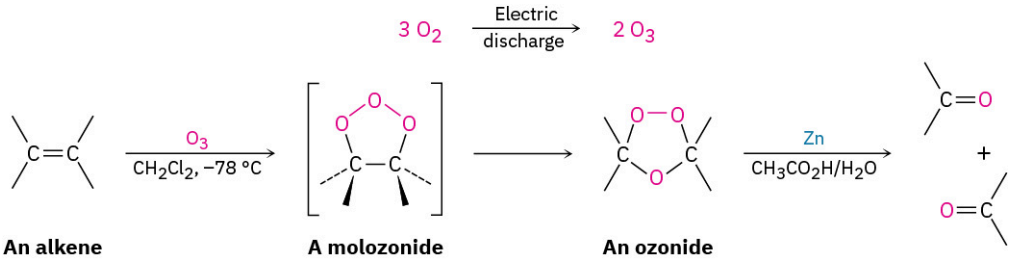
**(b)** 

**(c)** 

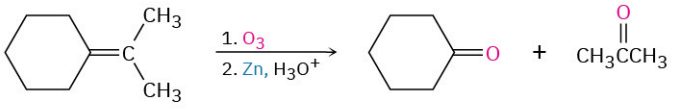
## 8.8 Oxidation of Alkenes: Cleavage to Carbonyl Compounds

In all the alkene addition reactions we've seen thus far, the carbon–carbon double bond has been converted into a single bond but the carbon skeleton has been unchanged. There are, however, powerful oxidizing reagents that will cleave C=C bonds and produce two carbonyl-containing fragments.

Ozone ($O_3$) is perhaps the most useful double-bond cleavage reagent. Prepared by passing a stream of oxygen through a high-voltage electrical discharge, ozone adds rapidly to a C=C bond at low temperature to give a cyclic intermediate called a *molozonide*. Once formed, the molozonide spontaneously rearranges to form an **ozonide**. Although we won't study the mechanism of this rearrangement in detail, it involves the molozonide coming apart into two fragments that then recombine in a different way.

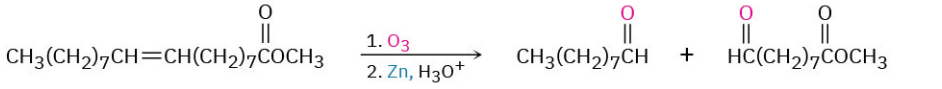



Low-molecular-weight ozonides are explosive and therefore not isolated. Instead, the ozonide is immediately treated with a reducing agent, such as zinc metal in acetic acid, to produce carbonyl compounds. The net result of the ozonolysis/reduction sequence is that the C=C bond is cleaved and an oxygen atom becomes doubly bonded to each of the original alkene carbons. If an alkene with a tetrasubstituted double bond is ozonized, two ketone fragments result; if an alkene with a trisubstituted double bond is ozonized, one ketone and one aldehyde result; and so on.




**Isopropylidenecyclohexane (tetrasubstituted)** — **Cyclohexanone** + **Acetone**

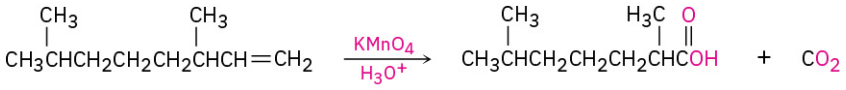
**84%; two ketones**




**Methyl 9-octadecenoate (disubstituted)** — **Nonanal** + **Methyl 9-oxononanoate**

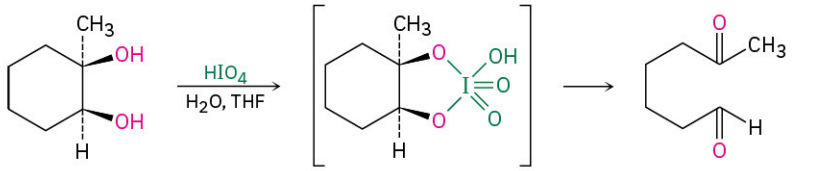
**78%; two aldehydes**

Several oxidizing reagents other than ozone also cause double-bond cleavage, although such reactions are not often used. For example, potassium permanganate ($KMnO_4$) in neutral or acidic solution cleaves alkenes to give carbonyl-containing products. If hydrogens are present on the double bond, carboxylic acids are produced; if two hydrogens are present on one carbon, $CO_2$ is formed.

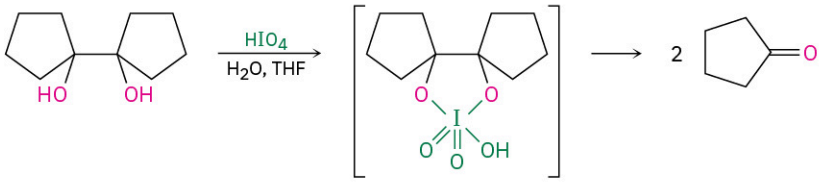



**3,7-Dimethyl-1-octene** — **2,6-Dimethylheptanoic acid (45%)**

In addition to direct cleavage with ozone or $KMnO_4$, an alkene can also be cleaved in a two-step process by initial hydroxylation to a 1,2-diol, as discussed in the previous section, followed by treatment of the diol with periodic acid, $HIO_4$. If the two –OH groups are in an open chain, two carbonyl compounds result. If the two –OH groups are on a ring, a single, open-chain dicarbonyl compound is formed. As indicated in the following examples, the cleavage reaction takes place through a cyclic periodate intermediate.




**A 1,2-diol** — **Cyclic periodate intermediate** — **6-Oxoheptanal (86%)**

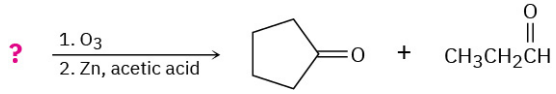



**A 1,2-diol** — **Cyclic periodate intermediate** — **Cyclopentanone (81%)**

## WORKED EXAMPLE 8.3

### Predicting the Reactant in an Ozonolysis Reaction

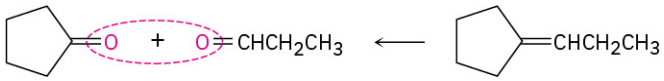
What alkene would yield a mixture of cyclopentanone and propanal on treatment with ozone followed by reduction with zinc?



**Strategy**

Reaction of an alkene with ozone, followed by reduction with zinc, cleaves the C=C bond and becomes two C=O bonds. Working backward from the carbonyl-containing products, the alkene precursor can be found by removing the oxygen from each product and joining the two carbon atoms.

**Solution**




**PROBLEM 8-15** What products would you expect from reaction of 1-methylcyclohexene with the following reagents?

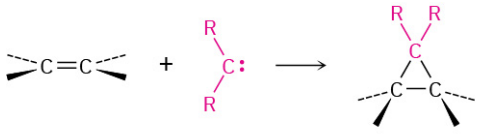
**(a)** Aqueous acidic $KMnO_4$ **(b)** $O_3$, followed by Zn, $CH_3CO_2H$

**PROBLEM 8-16** Propose structures for alkenes that yield the following products on reaction with ozone followed by treatment with Zn:

**(a)** $(CH_3)_2C{=}O + H_2C{=}O$ **(b)** 2 equiv $CH_3CH_2CH{=}O$

## 8.9 Addition of Carbenes to Alkenes: Cyclopropane Synthesis

Yet another kind of alkene addition is the reaction with a *carbene* to yield a cyclopropane. A **carbene, $R_2C$:**, is a neutral molecule containing a divalent carbon with only six electrons in its valence shell. It is therefore highly reactive and generated only as a reaction intermediate, rather than as an isolable molecule. Because they're electron-deficient, carbenes behave as electrophiles and react with nucleophilic C=C bonds. The reaction occurs in a single step without intermediates.

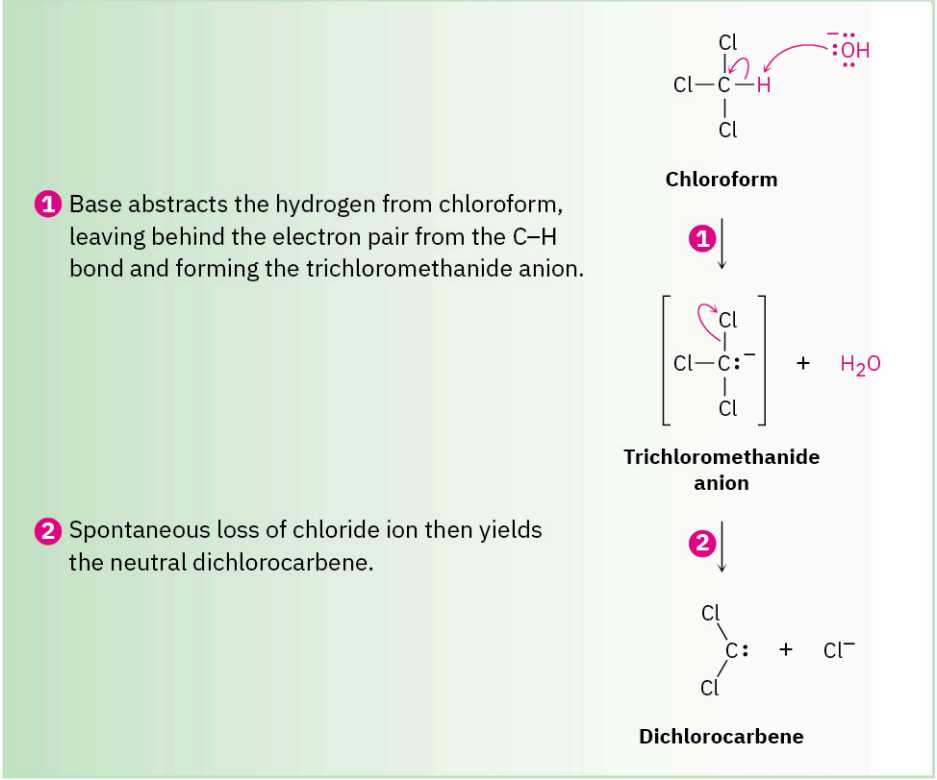



**An alkene** **A carbene** **A cyclopropane**

One of the simplest methods for generating a substituted carbene is by treatment of chloroform, $CHCl_3$, with a strong base such as KOH. As shown in **FIGURE 8.10**, the loss of a proton from $CHCl_3$ gives trichloromethanide anion, $^{-}{:}CCl_3$, which spontaneously expels a $Cl^-$ ion to yield dichlorocarbene, $:CCl_2$.

**FIGURE 8.10 MECHANISM**

**Mechanism of the formation of dichlorocarbene by reaction of chloroform with strong base.** Deprotonation of $CHCl_3$ gives the trichloromethanide anion, $^{-}:CCl_3$, which spontaneously expels a $Cl^-$ ion.




The carbon atom in dichlorocarbene is $sp^2$-hybridized, with a vacant $p$ orbital extending above and below the plane of the three atoms and with an unshared pair of electrons occupying the third $sp^2$ lobe. Note that this electronic description of dichlorocarbene is similar to that of a carbocation (Section 7.9) with respect to both the $sp^2$ hybridization of carbon and the vacant $p$ orbital. Electrostatic potential maps further illustrate the similarity (FIGURE 8.11).

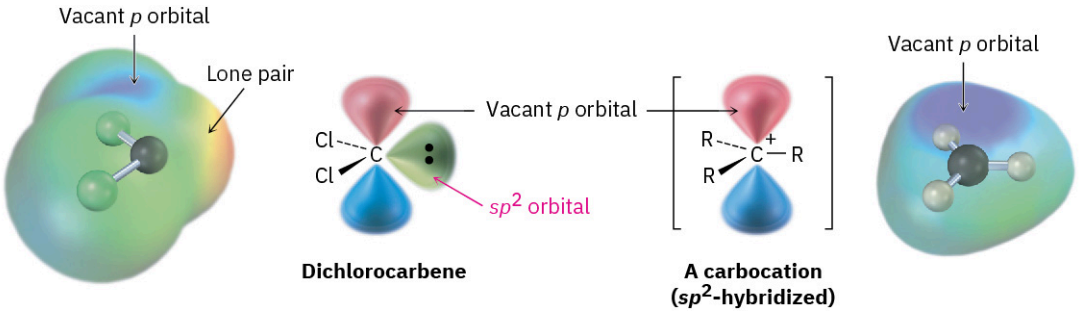
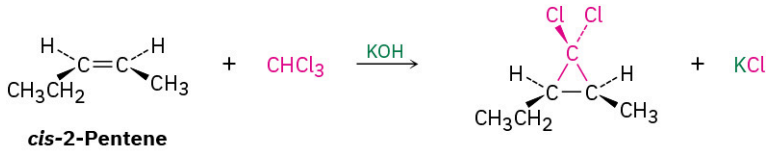
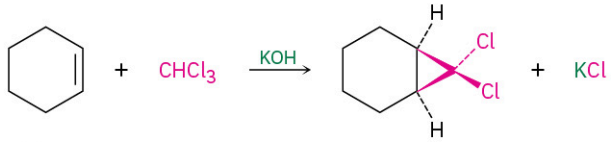



**FIGURE 8.11 The structure of dichlorocarbene.** Electrostatic potential maps show how **the positive region** coincides with the empty $p$ orbital in both dichlorocarbene and a carbocation ($CH_3^+$). The **negative region** in the dichlorocarbene map coincides with the lone-pair electrons.

If dichlorocarbene is generated in the presence of an alkene, addition to the double bond occurs and a dichlorocyclopropane is formed. As the reaction of dichlorocarbene with *cis*-2-pentene demonstrates, the addition is **stereospecific**, meaning that only a single stereoisomer is formed as product. Starting from a cis alkene, for instance, only cis-disubstituted cyclopropane is produced; starting from a trans alkene, only trans-disubstituted cyclopropane is produced.

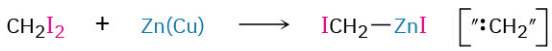



*cis*-2-Pentene

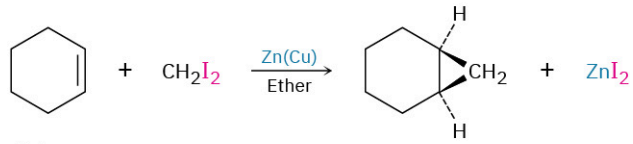



Cyclohexene

The best method for preparing nonhalogenated cyclopropanes is by a process called the **Simmons–Smith reaction**. First investigated at the DuPont company, this reaction does not involve a free carbene. Rather, it utilizes a *carbenoid*—a metal-complexed reagent with carbene-like reactivity. When diiodomethane is treated with a specially prepared zinc–copper mix, (iodomethyl)zinc iodide, $ICH_2ZnI$, is formed. In the presence of an alkene, $ICH_2ZnI$ transfers a $CH_2$ group to the double bond to yield cyclopropane. For example, cyclohexene reacts cleanly and with good yield to give the corresponding cyclopropane. Although we won't discuss the mechanistic details, carbene addition to an alkene is one of a general class of reactions called *cycloadditions,* which we'll study more carefully in Chapter 30.



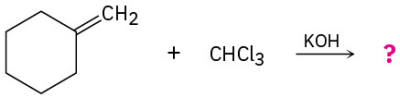

Diiodomethane (Iodomethyl)zinc iodide (a carbenoid)



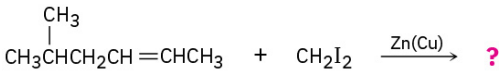

Cyclohexene Bicyclo[4.1.0]heptane (92%)

**PROBLEM 8-17** What products would you expect from the following reactions?

**(a)**




**(b)**

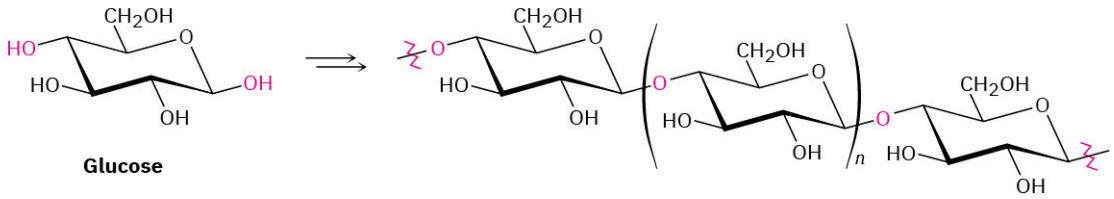



## 8.10 Radical Additions to Alkenes: Chain-Growth Polymers

In our brief introduction to radical reactions in Section 6.6, we said that radicals can add to C=C bonds, taking one electron from the double bond and leaving one behind to yield a new radical. Let's now look at the process in more detail, focusing on the industrial synthesis of alkene polymers. A **polymer** is a large—sometimes *very* large—molecule, built up by repetitive joining together of many smaller molecules, called **monomers**.

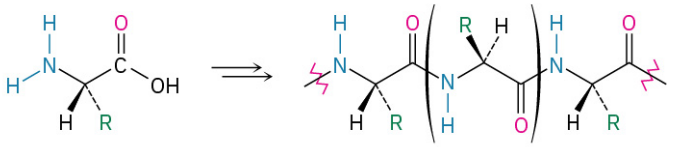
Nature makes wide use of biological polymers. Cellulose, for instance, is a polymer built of repeating glucose monomer units; proteins are polymers built of repeating amino acid monomers; and nucleic acids are polymers built of repeating nucleotide monomers.

**Cellulose—a glucose polymer**



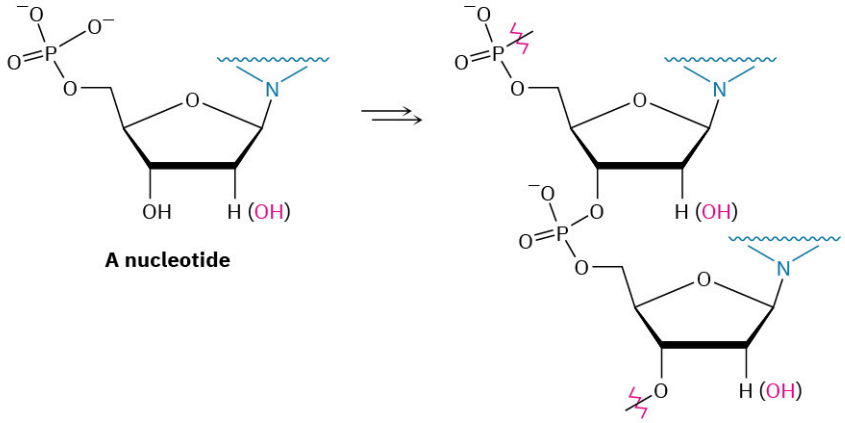

Glucose Cellulose

**Protein—an amino acid polymer**



**An amino acid** **A protein**

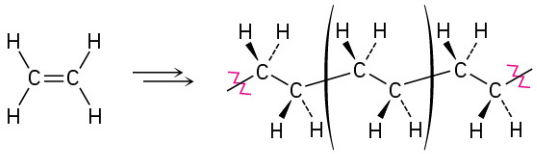
**Nucleic acid—a nucleotide polymer**



**A nucleotide** **A nucleic acid**

Synthetic polymers, such as polyethylene, are much simpler chemically than biopolymers, but there is still a great diversity to their structures and properties, depending on the identity of the monomers and on the reaction conditions used for polymerization. The simplest synthetic polymers are those that result when an alkene is treated with a small amount of a suitable catalyst. Ethylene, for example, yields polyethylene, an enormous alkane that may have a molecular weight up to *6 million* u and may contain as many as 200,000 monomer units. Worldwide production of polyethylene is approximately 88 million tons per year.

**Polyethylene—a synthetic alkene polymer**

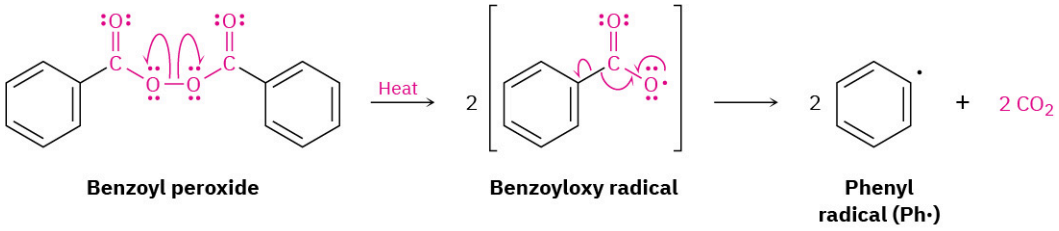


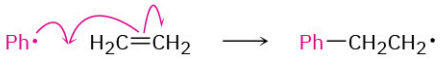
**Ethylene** **Polyethylene**

Polyethylene and other simple alkene polymers are called **chain-growth polymers** because they are formed in a chain-reaction process in which an initiator adds to a carbon–carbon double bond to yield a reactive intermediate. The intermediate then reacts with a second molecule of monomer to yield a new intermediate, which reacts with a third monomer unit, and so on.

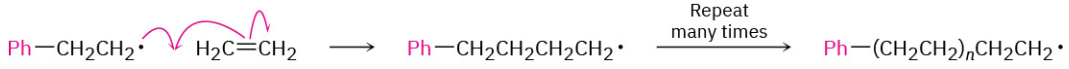
Historically, ethylene polymerization was carried out at high pressure (1000–3000 atm) and high temperature (100–250 °C) in the presence of a radical initiator such as benzoyl peroxide. Like many radical reactions, the mechanism of ethylene polymerization occurs in three steps: initiation, propagation, and termination:

- **Initiation** The polymerization reaction is initiated when a few radicals are generated on heating a small amount of benzoyl peroxide catalyst to break the weak O–O bond. The initially formed benzoyloxy radical loses $CO_2$ and gives a phenyl radical (Ph·), which adds to the C=C bond of ethylene to start the polymerization process. One electron from the ethylene double bond pairs up with the odd electron on the phenyl radical to form a new C–C bond, and the other electron remains on carbon.



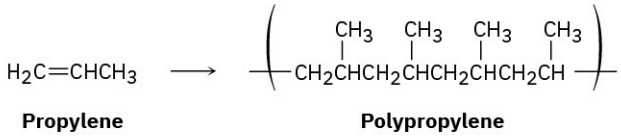

- **Propagation** Polymerization occurs when the carbon radical formed in the initiation step adds to another ethylene molecule to yield another radical. Repetition of the process for hundreds or thousands of times builds the polymer chain.



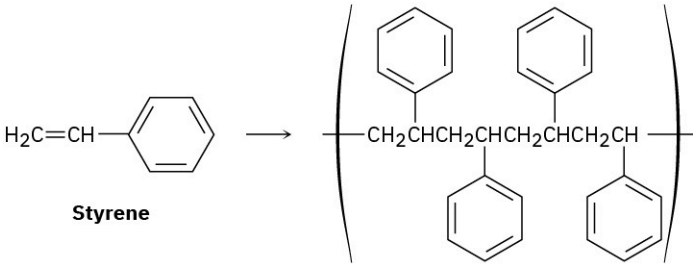

- **Termination** The chain process is eventually ended by a reaction that consumes the radical. The combination of two growing chains is one possible chain-terminating reaction.

$$2\,R\text{–}CH_2CH_2\cdot \longrightarrow R\text{–}CH_2CH_2CH_2CH_2\text{–}R$$

Ethylene is not unique in its ability to form a polymer. Many substituted ethylenes, called vinyl monomers, also undergo polymerization to yield polymers with substituent groups regularly spaced on alternating carbon atoms along the chain. Propylene, for example, yields polypropylene, and styrene yields polystyrene.



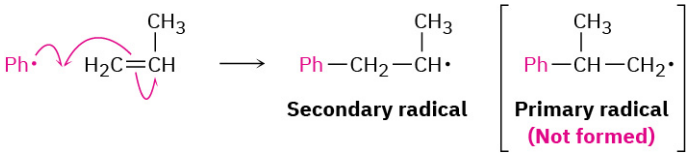

**Propylene** **Polypropylene**




**Styrene**

**Polystyrene**

When an unsymmetrically substituted vinyl monomer such as propylene or styrene is polymerized, the radical addition steps can take place at either end of the double bond to yield either a primary radical intermediate ($RCH_2\cdot$) or a secondary radical ($R_2CH\cdot$). Just as in electrophilic addition reactions, however, we find that only the more highly substituted, secondary radical is formed.




**Secondary radical** **Primary radical (Not formed)**

TABLE 8.1 shows some commercially important alkene polymers, their uses, and the monomers from which they are made.

TABLE 8.1 Some Alkene Polymers and Their Uses

| Monomer | Formula | Trade or common name of polymer | Uses |
|---|---|---|---|
| Ethylene | $H_2C{=}CH_2$ | Polyethylene | Packaging, bottles |
| Propene (propylene) | $H_2C{=}CHCH_3$ | Polypropylene | Moldings, rope, carpets |
| Chloroethylene (vinyl chloride) | $H_2C{=}CHCl$ | Poly(vinyl chloride) | Insulation, films, pipes |
| Styrene | $H_2C{=}CHC_6H_5$ | Polystyrene | Foam, moldings |
| Tetrafluoroethylene | $F_2C{=}CF_2$ | Teflon | Gaskets, nonstick coatings |
| Acrylonitrile | $H_2C{=}CHCN$ | Orlon, Acrilan | Fibers |
| Methyl methacrylate | $H_2C{=}C(CH_3)CO_2CH_3$ | Plexiglas, Lucite | Paint, sheets, moldings |
| Vinyl acetate | $H_2C{=}CHOCOCH_3$ | Poly(vinyl acetate) | Paint, adhesives, foams |

## WORKED EXAMPLE 8.4

### Predicting the Structure of a Polymer

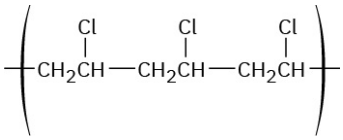
Show the structure of poly(vinyl chloride), a polymer made from $H_2C{=}CHCl$, by drawing several repeating units.

**Strategy**

Mentally break the carbon–carbon double bond in the monomer unit, and form single bonds by connecting numerous units together.

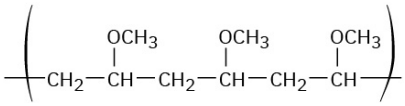
**Solution**

The general structure of poly(vinyl chloride) is

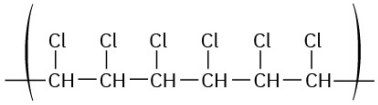


**PROBLEM 8-18** Show the monomer units you would use to prepare the following polymers:

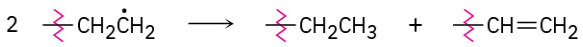
**(a)**



**(b)**



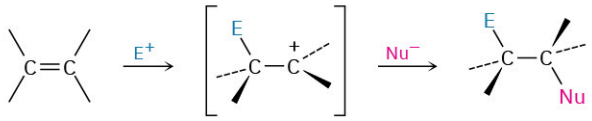
**PROBLEM 8-19** One of the chain-termination steps that sometimes occurs to interrupt polymerization is the following reaction between two radicals. Propose a mechanism for the reaction, using fishhook arrows to indicate electron flow.



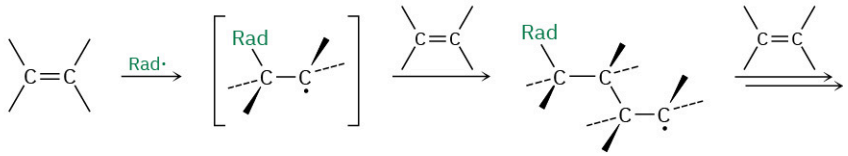
## 8.11 Biological Additions of Radicals to Alkenes

The same high reactivity of radicals that enables the alkene polymerization we saw in the previous section also makes it difficult to carry out controlled radical reactions on complex molecules. As a result, there are severe limitations on the usefulness of radical addition reactions in the laboratory. In contrast to an electrophilic addition, where reaction occurs once and the reactive cation intermediate is rapidly quenched by a nucleophile, the reactive intermediate in a radical reaction is not usually quenched. Instead, it reacts again and again in a largely uncontrollable way.

**Electrophilic addition**
**(Intermediate is quenched, so reaction stops.)**



**Radical addition**
**(Intermediate is not quenched, so reaction does not stop.)**



In biological reactions, the situation is different from that in the laboratory. Only one substrate molecule at a time is present in the active site of an enzyme, and that molecule is held in a precise position, with other necessary reacting groups nearby. As a result, biological radical reactions are more controlled and more common than laboratory or industrial radical reactions. A particularly impressive example occurs in the biosynthesis of prostaglandins from arachidonic acid, where a sequence of four radical additions take place. Its reaction mechanism was discussed briefly in **Section 6.6**.

As shown in **FIGURE 8.12**, prostaglandin biosynthesis begins with abstraction of a hydrogen atom from C13 of arachidonic acid by an iron–oxy radical to give a carbon radical that reacts with $O_2$ at C11 through a resonance form. The oxygen radical that results adds to the C8–C9 double bond to give a carbon radical at C8, which adds to the C12–C13 double bond and gives a carbon radical at C13. A resonance form of this carbon radical adds at C15 to a second $O_2$ molecule, completing the prostaglandin skeleton. Reduction of the O–O bond then gives prostaglandin $H_2$, called $PGH_2$. The pathway looks complicated, but the entire process is catalyzed with exquisite control by a single enzyme.

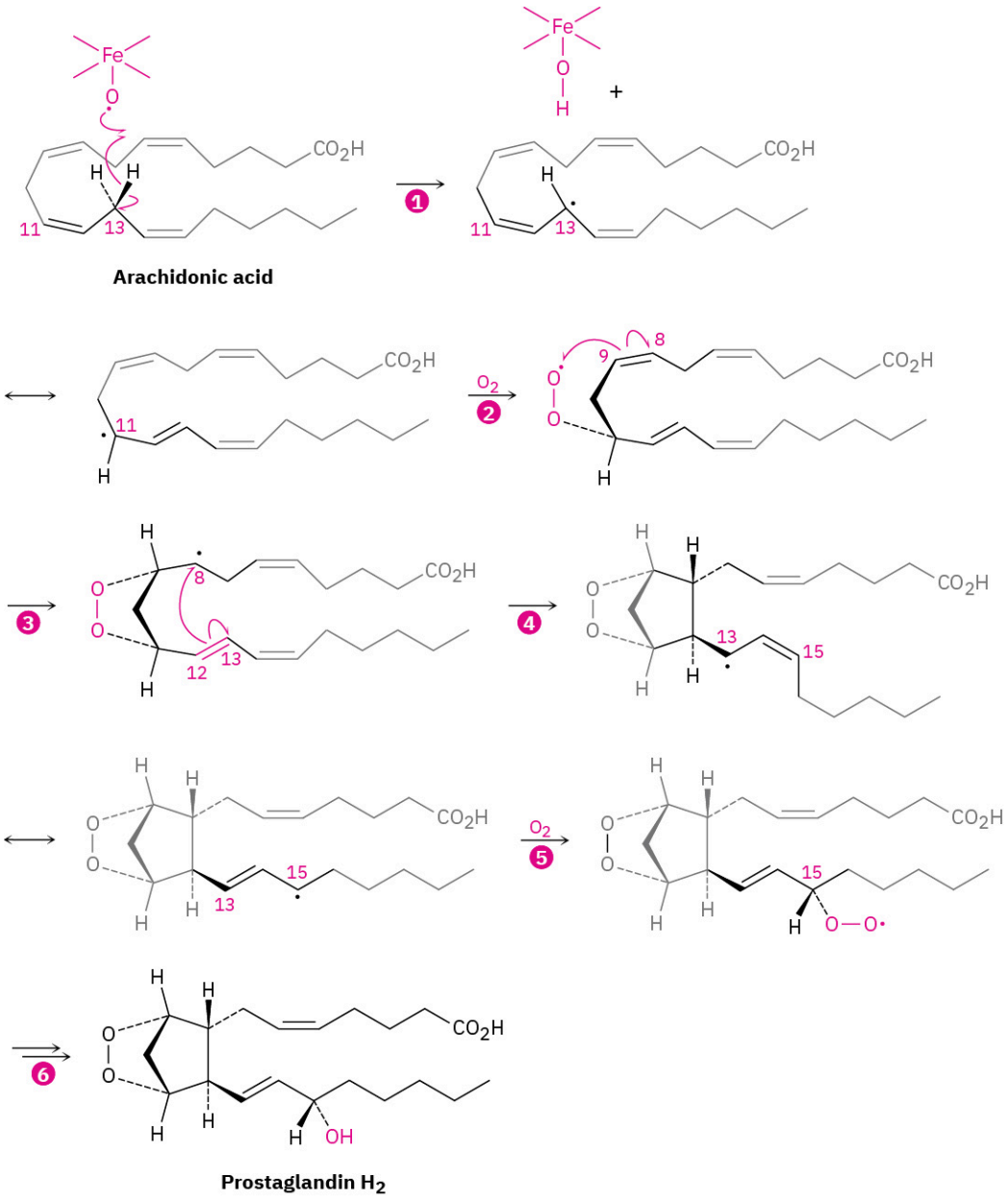
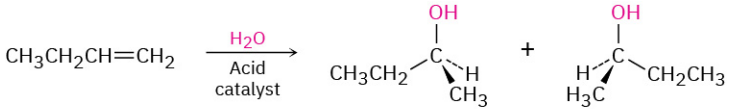



**FIGURE 8.12 Pathway for the biosynthesis of prostaglandins from arachidonic acid.** Steps **2** and **5** are radical addition reactions to $O_2$; steps **3** and **4** are radical additions to carbon–carbon double bonds.

## 8.12 Reaction Stereochemistry: Addition of $H_2O$ to an Achiral Alkene

Most of the biochemical reactions that take place in the body, as well as many organic reactions in the laboratory, yield products with chirality centers. For example, acid-catalyzed addition of $H_2O$ to 1-butene in the laboratory yields 2-butanol, a chiral alcohol. What is the stereochemistry of this chiral product? If a single enantiomer is formed, is it *R* or *S*? If a mixture of enantiomers is formed, how much of each? In fact, the 2-butanol produced is a racemic mixture of *R* and *S* enantiomers. Let's see why.




**1-Butene (achiral)** **(*S*)-2-Butanol (50%)** **(*R*)-2-Butanol (50%)**

To understand why a racemic product results from the reaction of $H_2O$ with 1-butene, think about the reaction

mechanism. 1-Butene is first protonated to yield an intermediate secondary carbocation. Because the trivalent carbon is $sp^2$-hybridized and planar, the cation has a plane of symmetry and is achiral. As a result, it can react with $H_2O$ equally well from either the top or the bottom. Reaction from the top leads to (*S*)-2-butanol through transition state 1 (TS 1) in **FIGURE 8.13**, and reaction from the bottom leads to (*R*)-2-butanol through TS 2. But the two transition states are mirror images, so they have identical energies, form at identical rates, and are equally likely to occur.

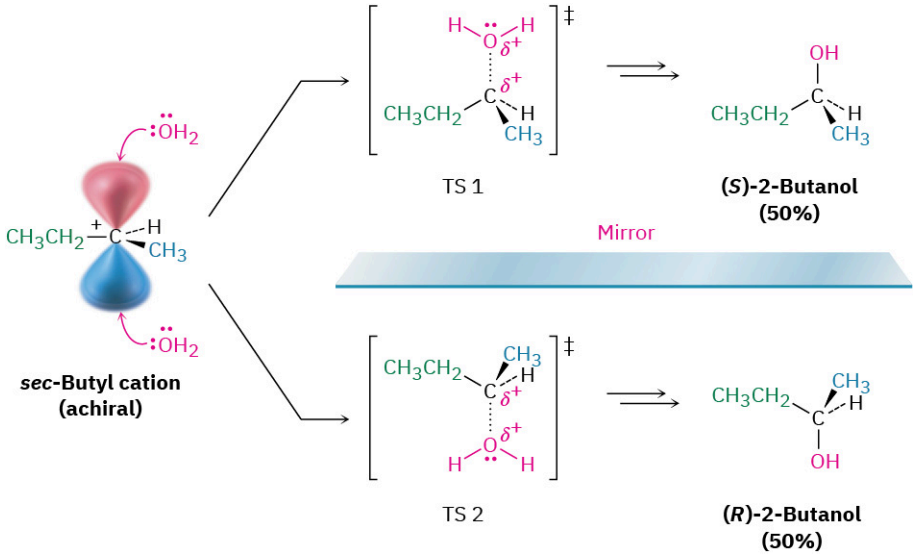
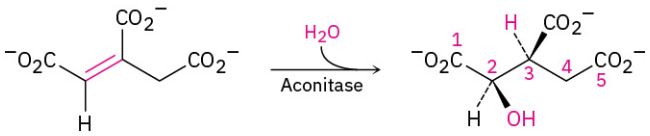



**FIGURE 8.13 Reaction of $H_2O$ with the carbocation resulting from protonation of 1-butene.** Reaction from the top leads to *S* product and is the mirror image of reaction from the bottom, which leads to *R* product. Because they are energetically identical, they are equally likely and lead to a racemic mixture of products. The dotted C···O bond in the transition state indicates partial bond formation.

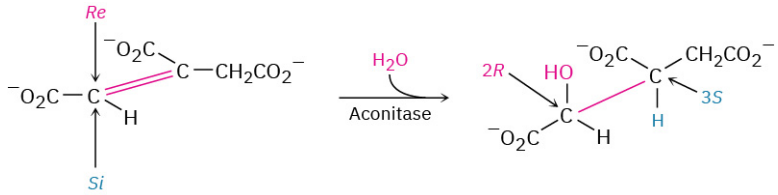
As a general rule, the formation of a new chirality center by achiral reactants always leads to a racemic mixture of enantiomeric products. Put another way, optical activity can't appear from nowhere; an optically active product can only result by starting with an optically active reactant or chiral environment (**Section 5.12**).

In contrast to laboratory reactions, enzyme-catalyzed biological reactions often give a single enantiomer of a chiral product, even when the substrate is achiral. One step in the citric acid cycle of food metabolism, for instance, is the aconitase-catalyzed addition of water to (*Z*)-aconitate (usually called *cis*-aconitate) to give isocitrate.




***cis*-Aconitate (achiral)** **(2*R*,3*S*)-Isocitrate**

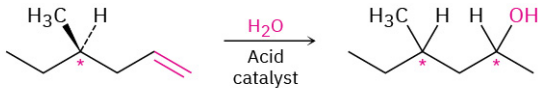
Even though *cis*-aconitate is achiral, only the (2*R*,3*S*) enantiomer of the product is formed. As discussed in **Section 5.11** and **Section 5.12**, *cis*-aconitate is a prochiral molecule, which is held in a chiral environment by the aconitase enzyme during the reaction. In this environment, the two faces of the double bond are chemically distinct, and addition occurs on only the Re face at C2.




***cis*-Aconitate** **(2*R*,3*S*)-Isocitrate**

## 8.13 Reaction Stereochemistry: Addition of $H_2O$ to a Chiral Alkene

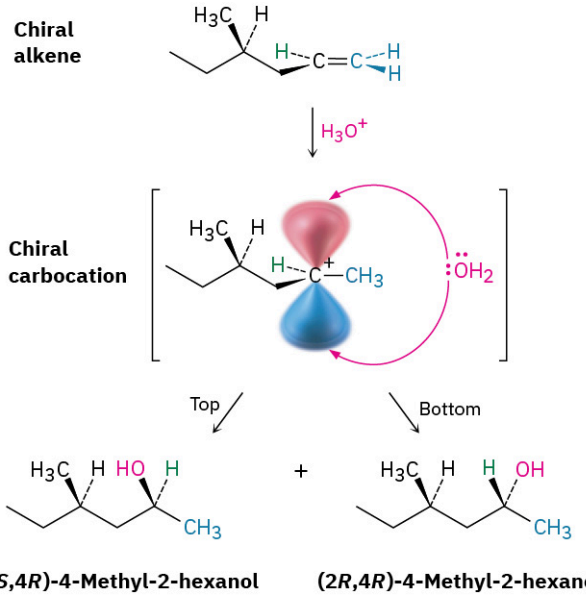
The reaction discussed in the previous section involves an addition to an achiral reactant and forms an optically inactive, racemic mixture of two enantiomeric products. What would happen, though, if we were to carry out the reaction on a single enantiomer of a chiral reactant? For example, what stereochemical result would be obtained from addition of $H_2O$ to a chiral alkene, such as (*R*)-4-methyl-1-hexene? The product of the reaction, 4-methyl-2-hexanol, has two chirality centers and so has four possible stereoisomers.




**(*R*)-4-Methyl-1-hexene (chiral)** **4-Methyl-2-hexanol (chiral)**

Let's think about the two chirality centers separately. What about the configuration at C4, the methyl-bearing carbon atom? Since C4 has the *R* configuration in the starting material and this chirality center is unaffected by the reaction, its configuration is unchanged. Thus, the configuration at C4 in the product remains *R* (assuming that the relative rankings of the four attached groups are not changed by the reaction).

What about the configuration at C2, the newly formed chirality center? As shown in **FIGURE 8.14**, the stereochemistry at C2 is established by reaction of $H_2O$ with a carbocation intermediate in the usual manner. But this carbocation doesn't have a plane of symmetry; it is chiral because of the chirality center at C4. Because the carbocation is chiral and has no plane of symmetry, it doesn't react equally well from the top and bottom faces. One of the two faces is likely, for steric reasons, to be a bit more accessible than the other, leading to a mixture of *R* and *S* products in some ratio other than 50 : 50. Thus, two diastereomeric products, (2*R*,4*R*)-4-methyl-2-hexanol and (2*S*,4*R*)-4-methyl-2-hexanol, are formed in unequal amounts, and the mixture is optically active.




**(2*S*,4*R*)-4-Methyl-2-hexanol** **(2*R*,4*R*)-4-Methyl-2-hexanol**

**FIGURE 8.14 Stereochemistry of the acid-catalyzed addition of $H_2O$ to the chiral alkene, (*R*)-4-methyl-1-hexene.** A mixture of diastereomeric 2*R*,4*R* and 2*S*,4*R* products is formed in unequal amounts because reaction of the chiral carbocation intermediate is not equally likely from top and bottom. The product mixture is optically active.

As a general rule, the formation of a new chirality center by a chiral reactant leads to unequal amounts of diastereomeric products. If the chiral reactant is optically active because only one enantiomer is used rather than a racemic mixture, then the products are also optically active.

**PROBLEM 8-20** What products are formed from acid-catalyzed hydration of racemic (±)-4-methyl-1-hexene? What can you say about the relative amounts of the products? Is the product mixture optically active?

**PROBLEM 8-21** What products are formed from hydration of 4-methylcyclopentene? What can you say about the relative amounts of the products?

# CHEMISTRY MATTERS

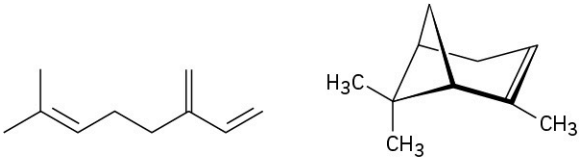
## Terpenes: Naturally Occurring Alkenes

Ever since its discovery in Persia around 1000 A.D., it has been known that *steam distillation*, the distillation of plant materials together with water, produces a fragrant mixture of liquids called essential oils. The resulting oils have long been used as medicines, spices, and perfumes, and their investigation played a major role in the emergence of organic chemistry as a science during the 19th century.



**FIGURE 8.15** The wonderful fragrance of leaves from the California bay laurel is due primarily to myrcene, a simple terpene. (credit: "California Bay Umbellularia californica" by Don Loaire/Flickr, CC BY 2.0)

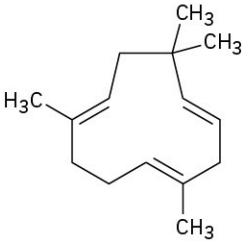
Chemically, plant essential oils consist largely of mixtures of compounds called terpenoids—small organic molecules with an immense diversity of structure. More than 60,000 different terpenoids are known. Some are open-chain molecules, and others contain rings; some are hydrocarbons, and others contain oxygen. Hydrocarbon terpenoids, in particular, are known as terpenes, and all contain double bonds. For example:



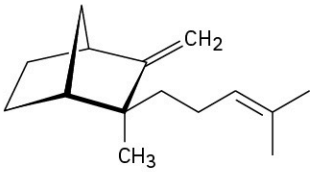
**Myrcene**
**(oil of bay)**




**α-Pinene**
**(turpentine)**



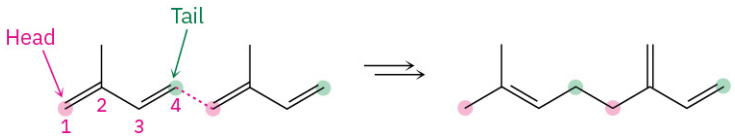

**Humulene**
**(oil of hops)**




**β-Santalene**
**(sandalwood oil)**

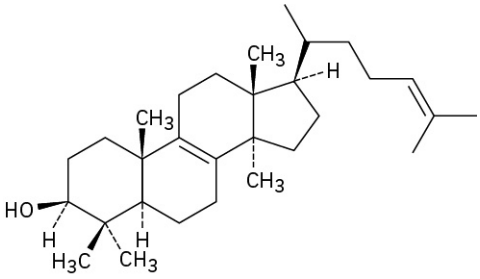
Regardless of their apparent structural differences, all terpenoids are related. According to a formalism called

the isoprene rule, they can be thought of as arising from head-to-tail joining of 5-carbon isoprene units (2-methyl-1,3-butadiene). Carbon 1 is the head of the isoprene unit, and carbon 4 is the tail. For example, myrcene contains two isoprene units joined head to tail, forming an 8-carbon chain with two 1-carbon branches. $\alpha$-Pinene similarly contains two isoprene units assembled into a more complex cyclic structure, and humulene contains three isoprene units. See if you can identify the isoprene units in $\alpha$-pinene, humulene, and $\beta$-santalene.



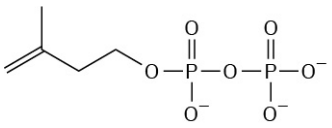

**Two isoprene units** **Myrcene**

Terpenes (and terpenoids) are further classified according to the number of 5-carbon units they contain. Thus, monoterpenes are 10-carbon substances derived from two isoprene units, sesquiterpenes are 15-carbon molecules derived from three isoprene units, diterpenes are 20-carbon substances derived from four isoprene units, and so on. Monoterpenes and sesquiterpenes are found primarily in plants, but the higher terpenoids occur in both plants and animals, and many have important biological roles. The triterpenoid lanosterol, for instance, is the biological precursor from which all steroid hormones are made.

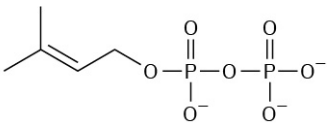



**Lanosterol**
**(a triterpene, $C_{30}$)**

Isoprene itself is not the true biological precursor of terpenoids. Nature instead uses two "isoprene equivalents"—isopentenyl diphosphate and dimethylallyl diphosphate—which are themselves made by two different routes depending on the organism. Lanosterol, in particular, is biosynthesized from acetic acid by a complex pathway that has been worked out in great detail. We'll look at the subject more closely in **Sections 27.5** and **27.7**.



**Isopentenyl diphosphate**



**Dimethylallyl diphosphate**

## Key Terms

- **anti stereochemistry**
- **bromonium ion**
- **carbene, $R_2C$**
- **chain-growth polymer**
- **epoxide**
- **glycol**
- **halohydrin**
- **hydroboration**
- **hydrogenated**
- **hydroxylation**
- **monomer**
- **oxidation**
- **oxirane**
- **oxymercuration–demercuration**
- **ozonide**
- **polymer**
- **reduction**
- **Simmons–Smith reaction**
- **stereospecific**
- **syn stereochemistry**

## Summary

With the background needed to understand organic reactions now covered, this chapter has begun the systematic description of major functional groups.

Alkenes are generally prepared by an elimination reaction, such as dehydrohalogenation, the elimination of HX from an alkyl halide, or dehydration, the elimination of water from an alcohol. The converse of this elimination reaction is the addition of various substances to the alkene double bond to give saturated products.

HCl, HBr, and HI add to alkenes by a two-step electrophilic addition mechanism. Initial reaction of the nucleophilic double bond with $H^+$ gives a carbocation intermediate, which then reacts with halide ion. Bromine and chlorine add to alkenes via three-membered-ring **bromonium ion** or chloronium ion intermediates to give addition products having **anti stereochemistry**. If water is present during the halogen addition reaction, a **halohydrin** is formed.

Hydration of an alkene—the addition of water—is carried out by either of two procedures, depending on the product desired. **Oxymercuration–demercuration** involves electrophilic addition of $Hg^{2+}$ to an alkene, followed by trapping of the cation intermediate with water and subsequent treatment with $NaBH_4$. **Hydroboration** involves addition of borane ($BH_3$) followed by oxidation of the intermediate organoborane with alkaline $H_2O_2$. The two hydration methods are complementary: oxymercuration–demercuration gives the product of Markovnikov addition, whereas hydroboration–oxidation gives the product with non-Markovnikov **syn stereochemistry**.

Alkenes are **reduced** by addition of $H_2$ in the presence of a catalyst such as platinum or palladium to yield alkanes, a process called catalytic **hydrogenation**. Alkenes are also **oxidized** by reaction with a peroxyacid to give **epoxides**, which can be converted into trans-1,2-diols by acid-catalyzed hydrolysis. The corresponding cis-1,2-diols can be made directly from alkenes by **hydroxylation** with $OsO_4$. Alkenes can also be cleaved to produce carbonyl compounds by reaction with ozone, followed by reduction with zinc metal. In addition, alkenes react with divalent substances called **carbenes, $R_2C$:**, to give cyclopropanes. Nonhalogenated cyclopropanes are best prepared by treatment of the alkene with $CH_2I_2$ and zinc–copper, a process called the **Simmons–Smith reaction**.

Alkene **polymers**—large molecules resulting from repetitive bonding of many hundreds or thousands of small **monomer** units—are formed by chain-reaction polymerization of simple alkenes. Polyethylene, polypropylene, and polystyrene are examples. As a general rule, radical addition reactions are not common in the laboratory but occur frequently in biological pathways.

Many reactions give chiral products. If the reactants are optically inactive, the products are also optically inactive. If one or both of the reactants is optically active, the products can also be optically active.

### LEARNING REACTIONS

What's seven times nine? Sixty-three, of course. You didn't have to stop and figure it out; you knew the answer immediately because you long ago learned the multiplication tables. Learning the reactions of organic chemistry requires the same approach: reactions have to be learned for quick recall if they are to be useful.

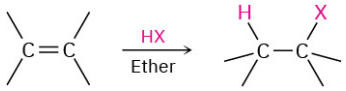
Different people take different approaches to learning reactions. Some people make flashcards; others find studying with friends to be helpful. To help guide your study, most chapters in this book end with a summary of the reactions just presented. In addition, the accompanying *Study Guide and Student Solutions Manual* has several appendixes that organize organic reactions from other perspectives. Fundamentally, though, there are no shortcuts. Learning organic chemistry does take effort.

## Summary of Reactions

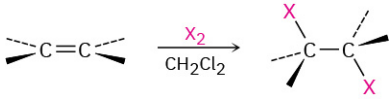
**No stereochemistry is implied unless specifically indicated with wedged, solid, and dashed lines.**

1. Addition reactions of alkenes

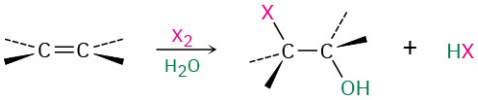
a. Addition of HCl, HBr, and HI (**Section 7.7** and **Section 7.8**)
Markovnikov regiochemistry occurs, with H adding to the less highly substituted alkene carbon and halogen adding to the more highly substituted carbon.



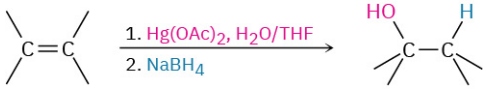

b. Addition of halogens $Cl_2$ and $Br_2$ (**Section 8.2**)
Anti addition is observed through a halonium ion intermediate.



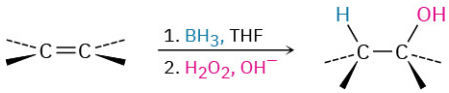

c. Halohydrin formation (**Section 8.3**)
Markovnikov regiochemistry and anti stereochemistry occur.



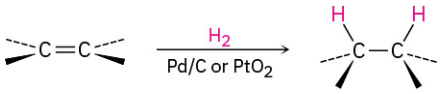

d. Addition of water by oxymercuration–demercuration (**Section 8.4**)
Markovnikov regiochemistry occurs.



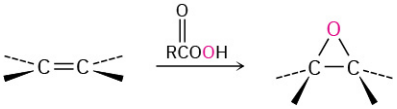

e. Addition of water by hydroboration–oxidation (**Section 8.5**)
Non-Markovnikov syn addition occurs.



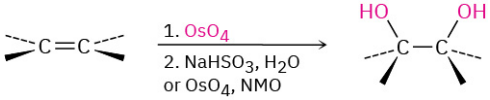

f. Catalytic hydrogenation (**Section 8.6**)
Syn addition occurs.



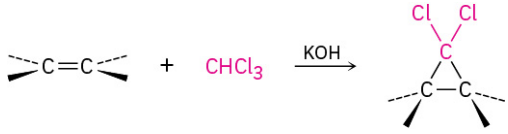

g. Epoxidation with a peroxyacid (**Section 8.7**)
Syn addition occurs.



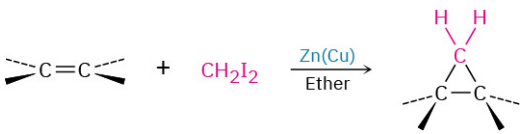

h. Hydroxylation with $OsO_4$ (**Section 8.7**)
Syn addition occurs.



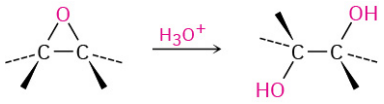

i. Addition of carbenes to yield cyclopropanes (**Section 8.9**)
(1) Dichlorocarbene addition



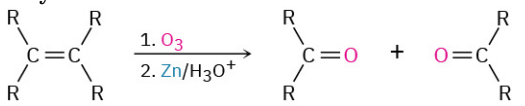

(2) Simmons–Smith reaction



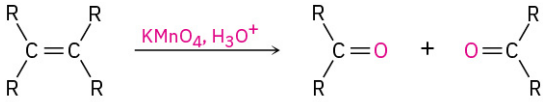
2. Hydroxylation by acid-catalyzed epoxide hydrolysis (Section 8.7)
   Anti stereochemistry occurs.

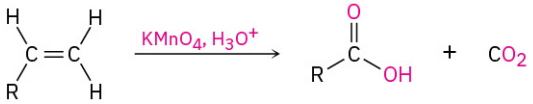



3. Oxidative cleavage of alkenes (Section 8.8)
   a. Reaction with ozone followed by zinc in acetic acid

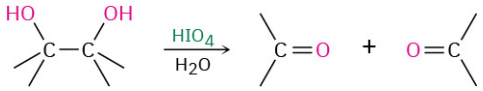



   b. Reaction with $KMnO_4$ in acidic solution




4. Cleavage of 1,2-diols (Section 8.8)

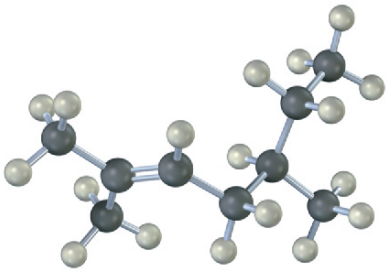



## Additional Problems

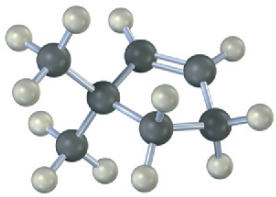
### Visualizing Chemistry

**PROBLEM 8-22** Name the following alkenes, and predict the products of their reaction with (1) *meta*-chloroperoxybenzoic acid, (2) $KMnO_4$ in aqueous acid, (3) $O_3$, followed by Zn in acetic acid:

**(a)**

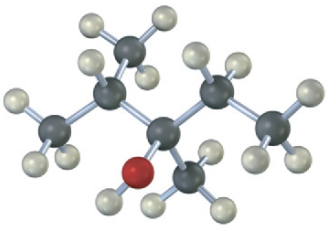


**(b)**

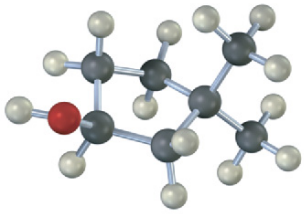


**PROBLEM 8-23** Draw the structures of alkenes that would yield the following alcohols on hydration (red = O). Tell in each case whether you would use hydroboration–oxidation or oxymercuration–demercuration.

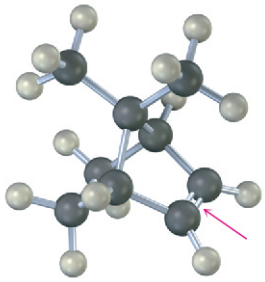
**(a)**



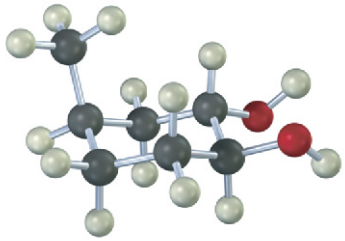
**(b)**



**PROBLEM 8-24** The following alkene undergoes hydroboration–oxidation to yield a single product rather than a mixture. Explain the result, and draw the product showing its stereochemistry.

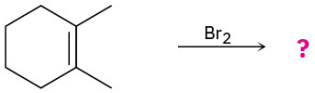


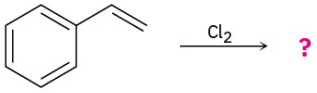
**PROBLEM 8-25** From what alkene was the following 1,2-diol made, and what method was used, epoxide hydrolysis or $OsO_4$?

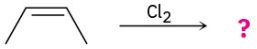


## Mechanism Problems

**PROBLEM 8-26** Predict the products for the following reactions, showing the complete mechanism and appropriate stereochemistry:

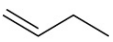
**(a)** 

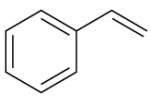

**(b)** 


**(c)** 

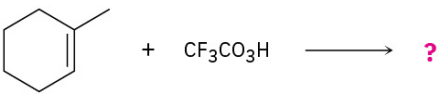

**PROBLEM 8-27** Draw the structures of the organoboranes formed when borane reacts with the following alkenes, including the regiochemistry and stereochemistry as appropriate. Propose a mechanism for each reaction.

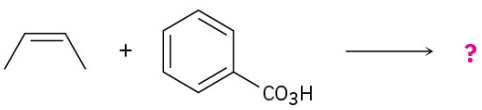
**(a)** 

**(b)** 

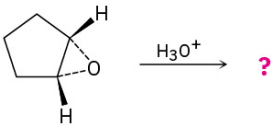
**(c)** 

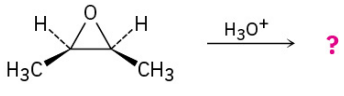
**PROBLEM 8-28** *meta*-Chloroperbenzoic acid is not the only peroxyacid capable of epoxide formation. For each reaction below, predict the products and show the mechanism.

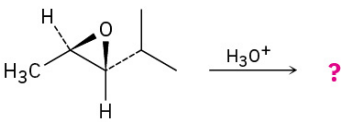
**(a)** 


**(b)** 


**PROBLEM 8-29** Give the mechanism and products for the following acid-catalyzed epoxide-opening reactions, including appropriate stereochemistry.

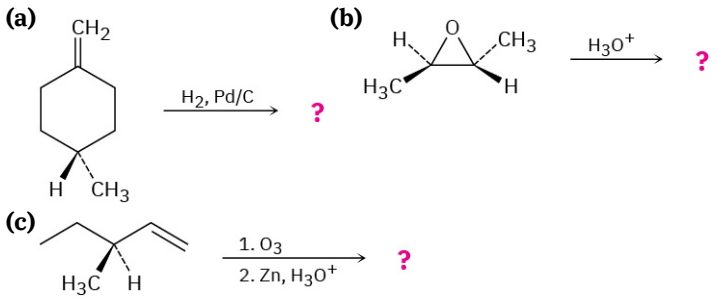
**(a)** 


**(b)** 


**(c)** 

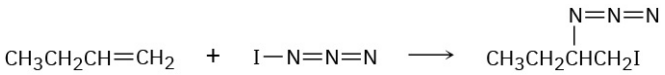

**PROBLEM** Which of the reactions below would result in a product mixture that would rotate plane-polarized

**8-30** light?




**PROBLEM 8-31** Reaction of 2-methylpropene with $CH_3OH$ in the presence of $H_2SO_4$ catalyst yields methyl *tert*-butyl ether, $CH_3OC(CH_3)_3$, by a mechanism analogous to that of acid-catalyzed alkene hydration. Write the mechanism, using curved arrows for each step.

**PROBLEM 8-32** Iodine azide, $IN_3$, adds to alkenes by an electrophilic mechanism similar to that of bromine. If a monosubstituted alkene such as 1-butene is used, only one product results:

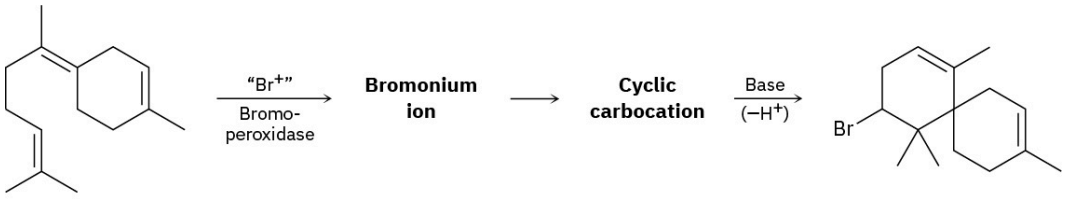



**(a)** Add lone-pair electrons to the structure shown for $IN_3$, and draw a second resonance form for the molecule.

**(b)** Calculate formal charges for the atoms in both resonance structures you drew for $IN_3$ in part **(a)**.

**(c)** In light of the result observed when $IN_3$ adds to 1-butene, what is the polarity of the $I-N_3$ bond? Propose a mechanism for the reaction using curved arrows to show the electron flow in each step.

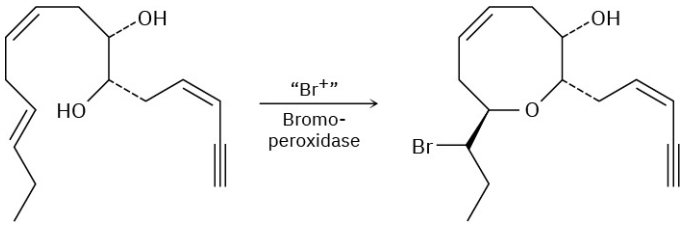
**PROBLEM 8-33** 10-Bromo-$\alpha$-chamigrene, a compound isolated from marine algae, is thought to be biosynthesized from $\gamma$-bisabolene by the following route:




**$\gamma$-Bisabolene** **10-Bromo-$\alpha$-chamigrene**

Draw the structures of the intermediate bromonium and cyclic carbocation, and propose mechanisms for all three steps.

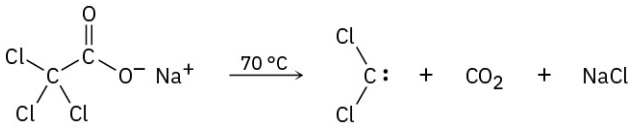
**PROBLEM 8-34** Isolated from marine algae, prelaureatin is thought to be biosynthesized from laurediol by the following route. Propose a mechanism.



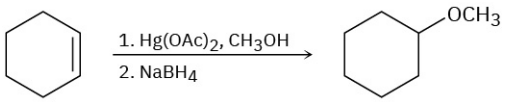

**Laurediol** **Prelaureatin**

**PROBLEM** Dichlorocarbene can be generated by heating sodium trichloroacetate. Propose a mechanism for

**8-35** the reaction, and use curved arrows to indicate the movement of electrons in each step. What relationship does your mechanism bear to the base-induced elimination of HCl from chloroform?

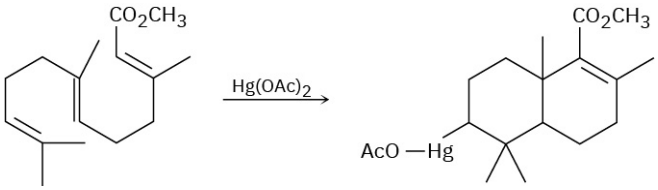



**PROBLEM 8-36** Reaction of cyclohexene with mercury(II) acetate in $CH_3OH$ rather than $H_2O$, followed by treatment with $NaBH_4$, yields cyclohexyl methyl ether rather than cyclohexanol. Suggest a mechanism.

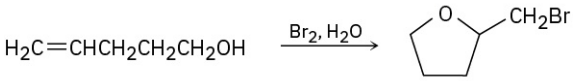



**Cyclohexene** **Cyclohexyl methyl ether**

**PROBLEM 8-37** Use your general knowledge of alkene chemistry to suggest a mechanism for the following reaction.

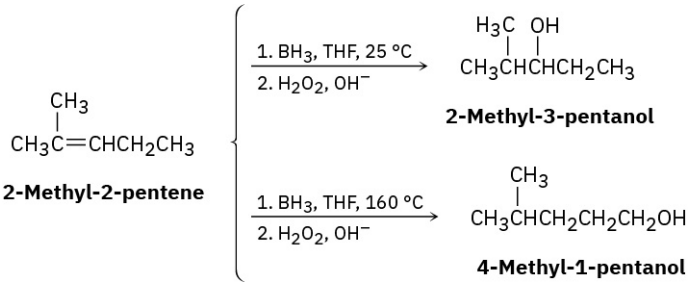



**PROBLEM 8-38** Treatment of 4-penten-1-ol with aqueous $Br_2$ yields a cyclic bromo ether rather than the expected bromohydrin. Suggest a mechanism, using curved arrows to show electron movement.




**4-Penten-1-ol** **2-(Bromomethyl)tetrahydrofuran**

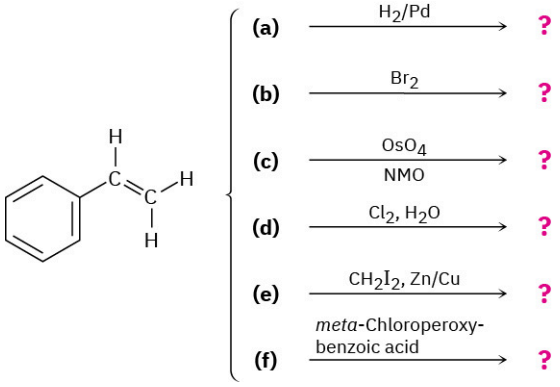
**PROBLEM 8-39** Hydroboration of 2-methyl-2-pentene at 25 °C, followed by oxidation with alkaline $H_2O_2$, yields 2-methyl-3-pentanol, but hydroboration at 160 °C followed by oxidation yields 4-methyl-1-pentanol. Suggest a mechanism.



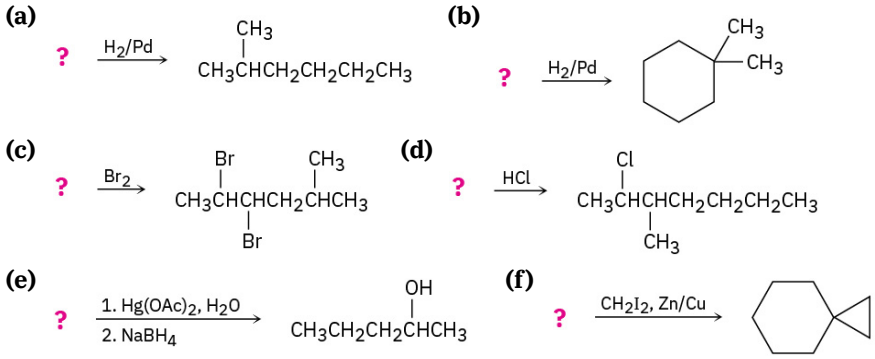

**2-Methyl-2-pentene** **2-Methyl-3-pentanol** **4-Methyl-1-pentanol**

## Reactions of Alkenes

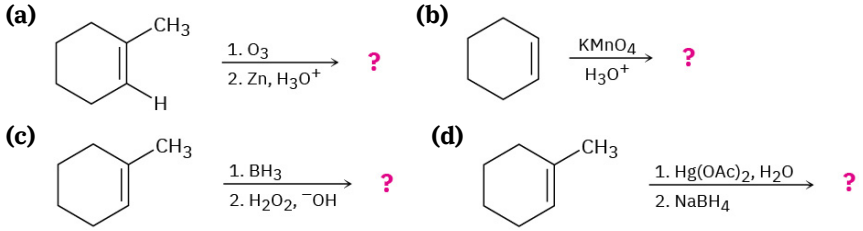
**PROBLEM 8-40** Predict the products of the following reactions (the aromatic ring is unreactive in all cases). Indicate regiochemistry when relevant.




**PROBLEM 8-41** Suggest structures for alkenes that give the following reaction products. There may be more than one answer for some cases.



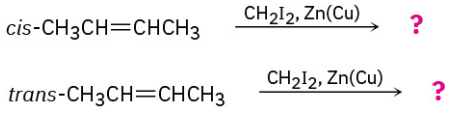

**PROBLEM 8-42** Predict the products of the following reactions, showing both regiochemistry and stereochemistry where appropriate:



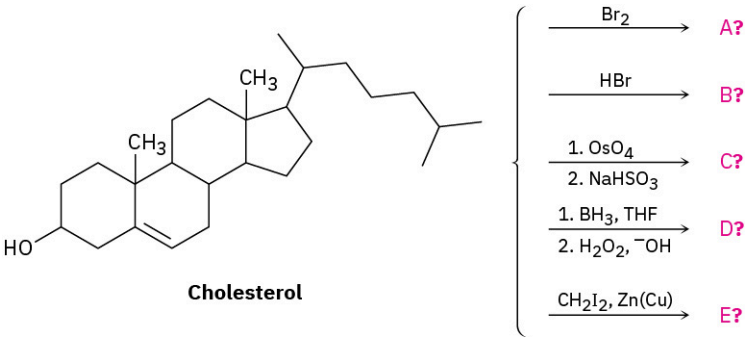

**PROBLEM 8-43** Which reaction would you expect to be faster, addition of HBr to cyclohexene or to 1-methylcyclohexene? Explain.

**PROBLEM 8-44** What product will result from hydroboration–oxidation of 1-methylcyclopentene with deuterated borane, $BD_3$? Show both the stereochemistry (spatial arrangement) and the regiochemistry (orientation) of the product.

**PROBLEM 8-45** The cis and trans isomers of 2-butene give different cyclopropane products in the Simmons–Smith reaction. Show the structures of both, and explain the difference.

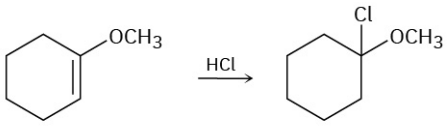



**PROBLEM 8-46** Predict the products of the following reactions. Don't worry about the size of the molecule; concentrate on the functional groups.




**Cholesterol**

**PROBLEM 8-47** Addition of HCl to 1-methoxycyclohexene yields 1-chloro-1-methoxycyclohexane as a sole product. Use resonance structures of the carbocation intermediate to explain why none of the alternate regioisomer is formed.

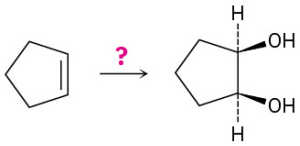



**1-Methoxycyclohexene** **1-Chloro-1-methoxy-cyclohexane**

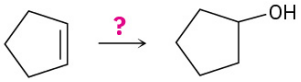
## Synthesis Using Alkenes

**PROBLEM 8-48** How would you carry out the following transformations? What reagents would you use in each case?

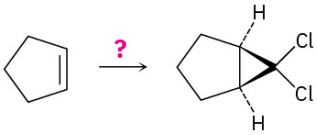
**(a)**



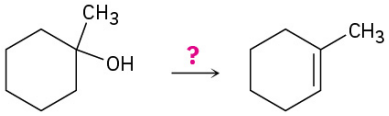

**(b)**



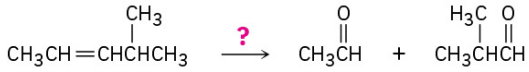

**(c)**




**(d)**




**(e)**




**(f)**




**PROBLEM 8-49** Draw the structure of an alkene that yields only acetone, $(CH_3)_2C{=}O$, on ozonolysis followed by treatment with Zn.

**PROBLEM 8-50** Show the structures of alkenes that give the following products on oxidative cleavage with $KMnO_4$ in acidic solution:

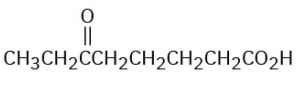
**(a)** $CH_3CH_2CO_2H$ + $CO_2$

**(b)** $(CH_3)_2C{=}O$ + $CH_3CH_2CH_2CO_2H$

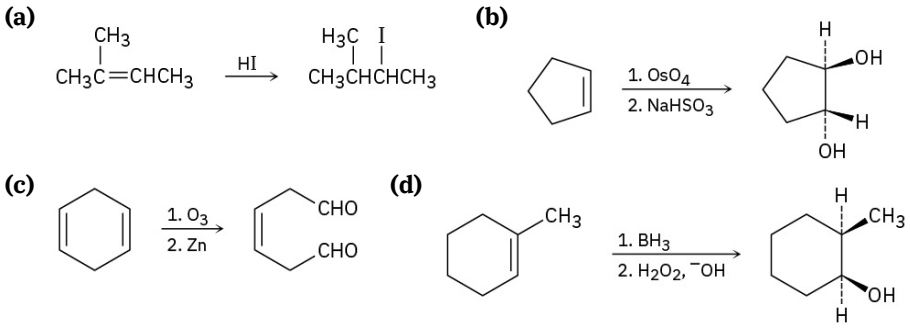
**(c)**



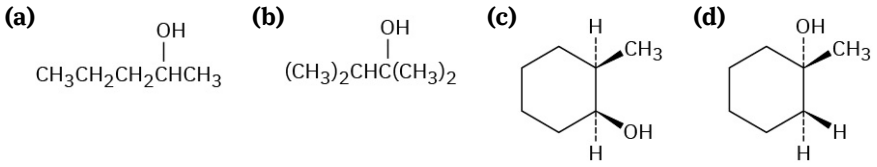

**(d)**




**PROBLEM 8-51** In planning the synthesis of one compound from another, it's just as important to know what not to do as to know what to do. The following reactions all have serious drawbacks to them. Explain the potential problems of each.

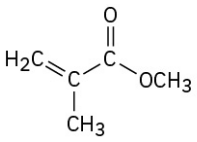



**PROBLEM 8-52** Which of the following alcohols could not be made selectively by hydroboration–oxidation of an alkene? Explain.



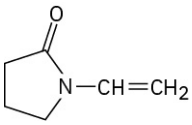

## Polymers

**PROBLEM 8-53** Plexiglas, a clear plastic used to make many molded articles, is made by polymerization of methyl methacrylate. Draw a representative segment of Plexiglas.



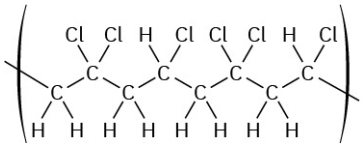

**Methyl methacrylate**

**PROBLEM 8-54** Poly(vinyl pyrrolidone), prepared from *N*-vinylpyrrolidone, is used both in cosmetics and as a component of a synthetic substitute for blood. Draw a representative segment of the polymer.




***N*-Vinylpyrrolidone**

**PROBLEM 8-55** When a single alkene monomer, such as ethylene, is polymerized, the product is a *homopolymer.* If a mixture of two alkene monomers is polymerized, however, a *copolymer* often results. The following structure represents a segment of a copolymer called *Saran.* What two monomers were copolymerized to make Saran?




**Saran**

## General Problems

**PROBLEM 8-56** Compound A has the formula $C_{10}H_{16}$. On catalytic hydrogenation over palladium, it reacts with only 1 molar equivalent of $H_2$. Compound **A** also undergoes reaction with ozone, followed by zinc treatment, to yield a symmetrical diketone, **B** ($C_{10}H_{16}O_2$).

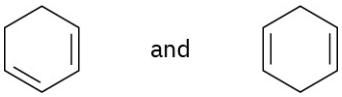
**(a)** How many rings does **A** have? **(b)** What are the structures of **A** and **B**?

**(c)** Write the reactions.

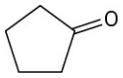
**PROBLEM 8-57** An unknown hydrocarbon **A** with the formula $C_6H_{12}$ reacts with 1 molar equivalent of $H_2$ over a palladium catalyst. Hydrocarbon **A** also reacts with $OsO_4$ to give diol **B**. When oxidized with $KMnO_4$

in acidic solution, **A** gives two fragments. One fragment is propanoic acid, $CH_3CH_2CO_2H$, and the other fragment is ketone **C**. What are the structures of **A**, **B**, and **C**? Write all reactions.

**PROBLEM 8-58** Using an oxidative cleavage reaction, explain how you would distinguish between the following two isomeric dienes:

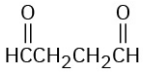



**PROBLEM 8-59** Compound **A**, $C_{10}H_{18}O$, undergoes reaction with dilute $H_2SO_4$ at 50 °C to yield a mixture of two alkenes, $C_{10}H_{16}$. The major alkene product, **B**, gives only cyclopentanone after ozone treatment followed by reduction with zinc in acetic acid. Identify **A** and **B**, and write the reactions.




**Cyclopentanone**

**PROBLEM 8-60** Draw the structure of a hydrocarbon that absorbs 2 molar equivalents of $H_2$ on catalytic hydrogenation and gives only butanedial on ozonolysis.




**Butanedial**

**PROBLEM 8-61** Simmons–Smith reaction of cyclohexene with diiodomethane gives a single cyclopropane product, but the analogous reaction of cyclohexene with 1,1-diiodoethane gives (in low yield) a mixture of two isomeric methylcyclopropane products. What are the two products, and how do they differ?

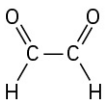
**PROBLEM 8-62** The sex attractant of the common housefly is a hydrocarbon with the formula $C_{23}H_{46}$. On treatment with aqueous acidic $KMnO_4$, two products are obtained, $CH_3(CH_2)_{12}CO_2H$ and $CH_3(CH_2)_7CO_2H$. Propose a structure.

**PROBLEM 8-63** Compound **A** has the formula $C_8H_8$. It reacts rapidly with $KMnO_4$ to give $CO_2$ and a carboxylic acid, **B** ($C_7H_6O_2$), but reacts with only 1 molar equivalent of $H_2$ on catalytic hydrogenation over a palladium catalyst. On hydrogenation under conditions that reduce aromatic rings, 4 equivalents of $H_2$ are taken up and hydrocarbon **C** ($C_8H_{16}$) is produced. What are the structures of **A**, **B**, and **C**? Write the reactions.

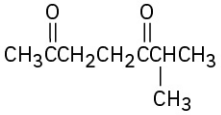
**PROBLEM 8-64** How would you distinguish between the following pairs of compounds using simple chemical tests? Tell what you would do and what you would see.

**(a)** Cyclopentene and cyclopentane **(b)** 2-Hexene and benzene

**PROBLEM 8-65** $\alpha$-Terpinene, $C_{10}H_{16}$, is a pleasant-smelling hydrocarbon that has been isolated from oil of marjoram. On hydrogenation over a palladium catalyst, $\alpha$-terpinene reacts with 2 molar equivalents of $H_2$ to yield a hydrocarbon, $C_{10}H_{20}$. On ozonolysis, followed by reduction with zinc and acetic acid, $\alpha$-terpinene yields two products, glyoxal and 6-methyl-2,5-heptanedione.




**Glyoxal**

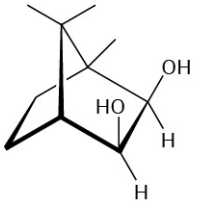



**6-Methyl-2,5-heptanedione**

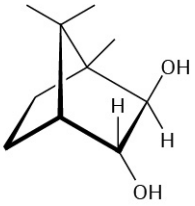
**(a)** How many degrees of unsaturation does $\alpha$-terpinene have?
**(b)** How many double bonds and how many rings does it have?
**(c)** Propose a structure for $\alpha$-terpinene.

**PROBLEM 8-66** Evidence that cleavage of 1,2-diols by $HIO_4$ occurs through a five-membered cyclic periodate intermediate is based on the measurement of reaction rates. When diols **A** and **B** were prepared and the rates of their reaction with $HIO_4$ were measured, it was found that diol **A** cleaved approximately 1 million times faster than diol **B**. Make molecular models of **A** and **B** and of potential cyclic

periodate intermediates, and then explain the results.

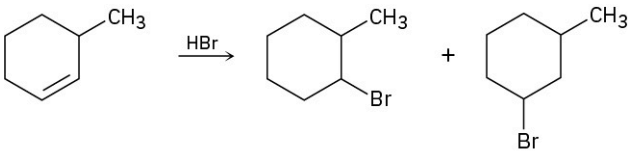



**A**
**(*cis* diol)**

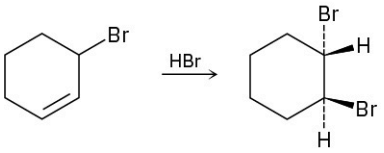



**B**
**(*trans* diol)**

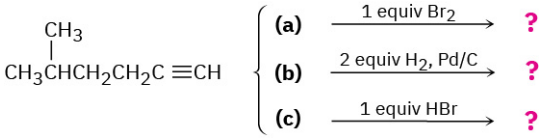
**PROBLEM 8-67** Reaction of HBr with 3-methylcyclohexene yields a mixture of four products: *cis*- and *trans*-1-bromo-3-methylcyclohexane and *cis*- and *trans*-1-bromo-2-methylcyclohexane. The analogous reaction of HBr with 3-bromocyclohexene yields *trans*-1,2-dibromocyclohexane as the sole product. Draw structures of the possible intermediates, and then explain why only a single product is formed in this reaction.




**cis, trans** **cis, trans**




**PROBLEM 8-68** We'll see in the next chapter that alkynes undergo many of the same reactions that alkenes do. What product might you expect from each of the following reactions?




**PROBLEM 8-69** Hydroxylation of *cis*-2-butene with $OsO_4$ yields a different product than hydroxylation of *trans*-2-butene. Draw the structure, show the stereochemistry of each product, and explain the difference between them.

**PROBLEM 8-70** Compound **A**, $C_{11}H_{16}O$, was found to be an optically active alcohol. Despite its apparent unsaturation, no hydrogen was absorbed on catalytic reduction over a palladium catalyst. On treatment of **A** with dilute sulfuric acid, dehydration occurred and an optically inactive alkene **B**, $C_{11}H_{14}$, was the major product. Alkene **B**, on ozonolysis, gave two products. One product was identified as propanal, $CH_3CH_2CHO$. Compound **C**, the other product, was shown to be a ketone, $C_8H_8O$. How many degrees of unsaturation does **A** have? Write the reactions, and identify **A**, **B**, and **C**.